

Towards Renewable energy targets for the Middle East and North African region: A decarbonization assessment of energy-water nexus

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ABSTRACT

Despite their huge fossil fuel potential, the Middle East and North Africa (MENA) region is sensitive to weakness in oil prices and this has a negative effect on the country's finances. Also, only 1.4% of the world's freshwater resources are found in MENA, despite having two-thirds or 66% of the world's known crude oil reserves. Despite that the energy importing countries in the MENA region benefit from lower energy prices, many of the countries like Libya, Syria, and Yemen have been shaken by wars, civil unrest, and political crisis, and this has created immense challenges in the energy situation. Another major challenge facing the MENA region is water scarcity. This region can be described as a desert area. There is a growing need for sustainable water management in the region as water scarcity is becoming increasingly critical. The government in North Africa have over the years set up laudable renewable energy projects to solve the prevailing energy crisis and improve their energy security. However, due to increasing population growth and increasing demand, there is still a huge gap between the renewable energy potential and generation that can be tapped. This study explores an hourly feasibility analysis of renewable energy production for electricity and freshwater production through the desalination process using the EnergyPLAN simulation environment. This study utilizes the planned renewable energy target set by the countries considered: Algeria, Morocco, Tunisia, Egypt, and Libya. The result shows that the increment of the renewable share to 52% in 2030 for Morocco will cause a reduction in carbon emission by 52.09%. The emission recorded for the 2030 scenario for 52% renewable energy integration was 14.209 Mt. This represents a CO₂ reduction of 78% from the 64.59 Mt emission in 2018. Also, for the projected 10% renewable electricity production target for Libya, the model shows that solar PV, wind energy, hydropower and CSP can supply a combined 6.61 TWh/year. The estimated technical result indicates that due to the extreme intermittency of renewable energy sources, the most probable scenario for meeting the future electricity demand in North Africa through a regional transmission grid is by hybrid renewable energy installations across the countries. Conclusively, it is presented that the actualization of renewable energy targets requires analysis and implementation of different policy strategies, especially for a region like North Africa, where there is a high reliance on fossil fuels for national revenue.

1. Introduction

The rate of progress of the global economy in meeting the sustainable development goals (SDG 7) is slow, and recent data has shown that the world is not within the perimeter of achieving the set targets (Barbier and Burgess, 2019). Even though there has been progress over the past ten years, it has been happening at a slower rate since it is difficult to reach the remaining population lacking electricity access. Around 2.4 billion people, or one-third of the world's population, still lacked access to clean cooking in 2020 as a result of population expansion exceeding

advancements in this area. Unless global efforts are increased, it is predicted that by 2030, 2.1 billion people will still lack access to clean cooking, and 670 million people will still lack access to power (Van Meijl et al., 2020). Some positive however is that despite the stalling of many energy projects in 2020, due to the COVID crisis, there was an 80% share of renewable energy in all new electricity capacity added in the year. The steady growth of renewable energy highlights the significant role it plays in cutting down on the global energy crisis (Tuan et al., 2021). However, the immediate fix of decarbonizing electricity production using renewable energy can bear down on more secondary societal

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challenges, and jeopardize sustainability goals (Oree et al., 2017). The consideration of certain factors like the water-energy-nexus for instance also needs to be accounted for, aside from the discourse of CO₂ reduction. The push for meeting renewable energy targets in different countries placed by international energy bodies can create competition for water, land, and food (Raina and Sinha, 2019). Decisions made today must take into account these effects since the energy infrastructure creates long-term path dependence. Making poor energy policy decisions and creating contradictory strategies might result in ignoring the relationship between energy, water, and land as well as the economic aspect of managing energy (Bazilian et al., 2011).

The energy situation in Africa is unique as compared to other parts of the world. This is so because of the socio-economic profile of the continent. Africa is the second-most populated continental region and is characterized by a high rate of population and income growth (Mandelli et al., 2014). Eastern, Middle, and Western Africa saw growth of about 2.5 per cent in 2010, with the average rate of growth for Africa according to the IEA being above 2.1 per cent. Comparatively, Europe 27 had growth at a pace of roughly 0.3%. Despite sub-Saharan Africa still having the highest death rate in the world, one of the primary causes of the population's rapid increase is the decline in child mortality rates. Furthermore, over the last decades, Africa's economy has been seen to experience fast economic growth, with forecasts showing that, on average, Africa's economy will outperform Asia's (Fund, 2019). For context, the comparison between Africa's and Europe 27 GDP shows that the African annual real GDP growth had been 5%, as against barely above 1% in Europe 27. However, despite the economic positives in the economic growth of Africa, which presents some positive basis for the discourse on present and future clean energy production in the region, some disparities are evident among the African countries and there is a large divide between urban and rural settlements, which breeds high poverty situations in large segments of the population (Gri, 2017). Ten countries in Africa, among which are 5 north African countries (Egypt, Algeria, Morocco, Tunisia, and Libya) contribute 75% of Africa's GDP, alongside, South Africa, Nigeria, Angola, Sudan and Ethiopia. The other 25% of Africa's GDP is supplied by the other countries, although their population is half of the continent (Mandelli et al., 2014), (Leibfritz and Flaig, 2013).

The Middle East and North Africa region is worth focusing on in the discourse of global energy crisis and renewable energy integration, as the region plays a vital part in the global energy system as a huge producer of fossil fuel, and is particularly susceptible to the socio-economic factors discussed in preceding paragraph. The MENA region is sensitive to weakness in oil prices, and hence has a negative effect on the country's finances. Also, despite that the energy importing countries in the MENA region benefit from lower energy prices, many of the countries like Libya, Syria, and Yemen have been shaken by wars, civil unrest, and political crisis. It is against the backdrop of diverse diplomatic relations in the region, and their relationship with the West, that energy policies in the region are being shaped (Hoppe et al., 2016) (Gri, 2017). The region is also endowed with natural resources like oil and gas (W. E. Council, 2013), with an estimation of about 860 billion barrels. This makes up about 52% of the total global oil reserves. The fossil fuel reserves of some North African countries are presented in Table 1. Also, the region is home to 42% of the total global natural gas reserves, with Saudi Arabia having the largest regional proved oil reserves. Although

the region contains more than 60% of the global oil reserves, there is a major challenge of fresh water scarcity.

All countries in the MENA region are subjected to scarcity of water, whereby renewable energy water resources are below 1000 m³/year/capita (Fitton et al., 2019). In a study by (Siddiqi and Anadon, 2011), the energy-water nexus in the MENA region was reviewed, and findings showed that there was a very substantial dependence on energy for fresh water. Their study mentioned that in Libya and Egypt, 14% of the total fuel consumption is due to groundwater pumping and wastewater treatment. This validates that the energy-water nexus is a key area for concern in MENA region.

This region can be described as a desert area. There is a growing need for sustainable water management in the region as water scarcity is becoming increasingly critical. For instance, groundwater-dependent ecosystems such as oases, and biodiversity hotspots, suffer from declining groundwater tables. North Africa is considered one of the driest regions in the world (Oweis and Hachum, 2018). The water scarcity in North Africa is an even more serious issue, as the Egyptian president, Abdel Fattah recently mentioned, that the country had reached the stage of water poverty, as the per capita water consumption is 500 cubic meters, which is way below the 1000 cubic meter per capita threshold level (Al-Monitor, 2022). A similar situation is prevalent in Libya. Ref (Brika, 2018) reported that Libya is one of the top countries with water stress in the world, as the country ranks as one of the most water-insecure places. The causes of water shortage in Libya are tied to factors like excessive groundwater exploitation, seawater intrusion, poor water management etc.

In solving the water crisis globally, the desalination process is a proven alternative water supply technology (Gude, 2017). The current worldwide desalination capacity is over 100 million cubic meters per day (MCM/d), which comes from approximately 16 thousand plants located in 175 countries across the globe. Roughly half of the global desalination capacity is held by nations in the Middle East and North Africa (MENA) (Jones et al., 2019). For instance, Morocco for over 30 years, has implemented salt water desalination technology, which has provided the country with about 20,000m³/day, representing 1.5% of the national drinking water production (Boye, 2008), (El-Ghizel et al., 2021). The challenge with the desalination process is that it is highly energy intensive. Presently, most of the desalination plants in the MENA region are run using fossil fuels, and this contributes to global warming. The nexus of energy and water is critical in creating an environmentally safe society, and also reducing water poverty. There is a growing interest in the MENA region regarding using renewable energy for the desalination process (Mahmoudi et al., 2009).

The research on energy transition shows that one of the major policy concerns facing industrial countries is the evolution of the energy system (Miller et al., 2013). This has been debated in the context of regional energy strategy and policy creation, and it is unquestionably true for MENA countries as well (Griffiths and Mills, 2016). Diversification of energy supply, management of energy demand, and energy commerce, particularly intra-regional trade, are crucial challenges for regional energy transformation. This is shown in Fig. 1, which explains the energy transition in the MENA region. This forms the core novelty of this study, especially considering the renewable energy developments in the region and the lofty renewable energy productions still being targeted. This study not only simulates the futuristic expected renewable energy productions based on estimated potential, but also discusses the energy trade, and regional policy situations. The futuristic implementation of renewable desalination in North Africa is also simulated using solar and wind energy technologies. In addition, according to the knowledge of the authors, this is the first study that utilizes an hourly-based model in simulating the synchronized renewable energy production for meeting future electricity demands in the North African region. This study also considers the present inter-regional grid connections among the country and the planned connections.

This study narrows down to the North African countries in the region

Table 1
Proved fossil fuel reserves in North Africa (Ghazi, 2021).

Country	Coal reserves (2017) Million tons	Crude oil reserves (2019) Billion barrels	Natural gas reserves (2019) Trillion m3
Algeria	65.04	12.2	159.05
Egypt	17.67	3.3	63
Morocco	15.43	0.00068	0.05
Tunisia	–	0.43	2.3

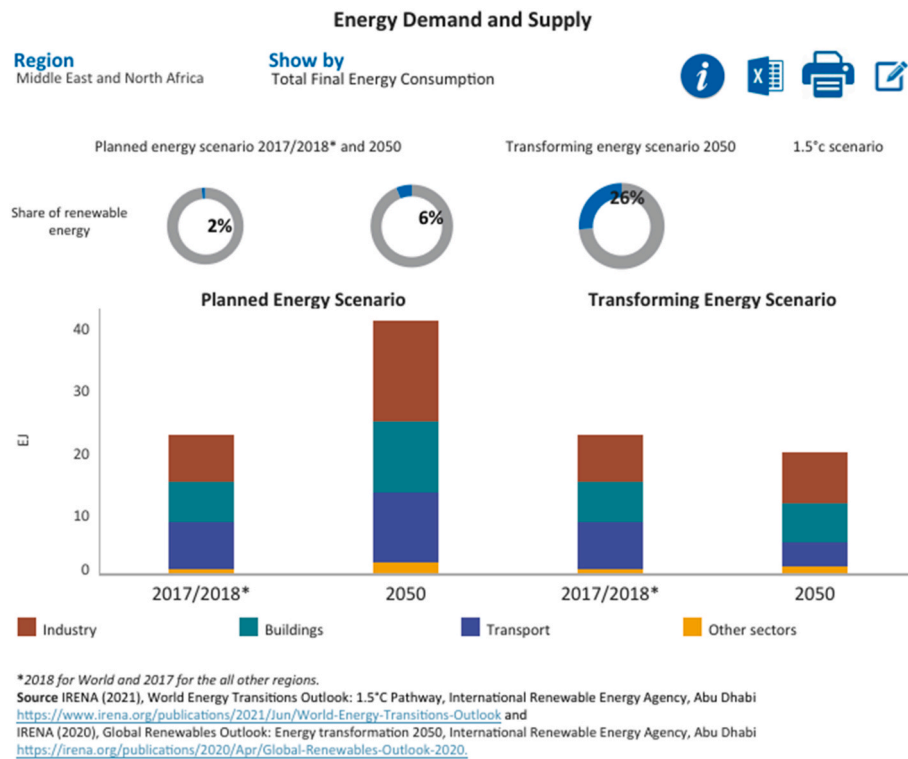


Fig. 1. Energy demand and supply in different sectors in the MENA region (IEA, 2020).

to allow for more concise discourse, however, a similar methodology can be transposed to the other countries in the MENA region. A study by Jebli et al. (Mehdi and Slim, 2017) studied the relationship between some econometric variables and renewable energy development in North African countries and concluded that renewable energy development is crucial for resolving the energy problems being faced in the region. A study by Diala et al. (Hawila et al., 2014a) carried out an assessment framework for investigating the renewable energy readiness of the North African countries. Their study identified gaps and strengths that are crucial for proposing RE developments in the countries. Table 2 shows studies that have modelled the technical and economic analysis of integrating or increasing the share of renewable energy in different countries using the EnergyPLAN software. This study builds on previous studies on RE development in North Africa, by analyzing the current and future state of the energy system in North Africa using the EnergyPLAN program. This study draws attention to the electricity supply in North Africa, and it is significant also from the perspective of the energy-water nexus considering that the existing water resource in the region is dependent on the electricity sector. Therefore, finding a viable route to decarbonizing the electricity sector would reduce carbon emissions relating to production of water and electricity. The goal of the authors therefore will be to discuss the technical and economic potential of RE integration in North Africa, as well as the policies that would favor their integration in the electricity grid. A discourse on the financing of RE technologies in North Africa is also presented.

Three scenarios were used in analyzing the decarbonization of the North African region. The first uses the Business as Usual (BAU) case, where the fossil fuel power plant is used in meeting future energy demands. The second scenario involves the integration of Renewable energy based on the projected RE share marked in the Country's RE future plan. The third scenario involves the inter-regional RE integration among the North African countries. The significance of this study is immense, as a clearer conversation in terms of the proposed renewable energy technologies in the region can be made. The objective of this paper is to model renewable energy integration for power generation and focus on desalination for freshwater production. The analysis made

in this study is built on certain conditions.

- The renewable energy integration targets set by the government in individual countries are used in the simulation process
- The simulation for each case study also included 100% renewable energy for electricity generation and the desalination process
- The simulation also considers regional cooperation and modelling of synchronous grid upon the assumption of a central grid across North Africa

The paper is sectioned to highlight significant review and analysis of the case study region. Section 2 contains the MENA region context, to show the definition of the region. Section 3 highlights the renewable energy potential of the MENA region, and North African countries studied in this research, Section 4 gives a concise relevant literature review of similar studies, section 5 and 6 contains the renewable energy targets and the methodology respectively. In section 7, the model results and discussion are made. Section 8 presents the discourse of financing RE technologies in MENA region, while section 9 gives the analysis of the intra-regional trade of renewable energy in the region. Section 10 highlights efficient policy formation needed for energy transition in the region, and the conclusion and recommendations is presented in section 11.

2. Definition of MENA region

The definition of the MENA region is not singular, due to different political, economic, and social sectioning, and divides. However, in this study, the World Bank definition of the region will be adopted. In this article, the MENA region is considered to be a conglomerate of North Africa (Algeria, Egypt, Libya, Morocco, and Tunisia), Mediterranean or Levant (Israel, Jordan, Lebanon, Palestine, Syria) Gulf Cooperation Council (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, United Arab Emirates) and Other Middle East (Iraq, Iran, Djibouti, Yemen), as depicted in Fig. 2.

As stated earlier, factors like social, political, and economic

Table 2
Some studies on energy system modelling using EnergyPLAN software.

Reference	Case study country	RE Targets and Objective	Methodology	Conclusions
Novosel et al. (Novosel et al., 2014)	Jordan	Increasing share of Renewable energy in electricity production and Desalination	Six scenarios were created using solar PV and wind energy until 2050	Their simulation result showed that renewable energy share can be increased up to 76%, from solar and wind technologies.
Connolly et al. (Connolly et al., 2011)	Ireland	Increasing the share of renewable energy to 100% in the electricity, heat, and transport sectors	Three scenarios were created using different renewable energy technology combinations	Their study stated that 100% RE integration in Ireland is not feasible. Additionally, their study mentioned that by deploying the ideal combination of technologies, significant reductions in the amount of energy used can be obtained with careful future planning.
Mario et al. (Pastore et al., 2022)	Italy	Reducing the natural gas consumption	Three scenarios were developed: the capacity of PV and wind energy was increased in steps of 2 GW, up to 5GW and 10 GW for PV and wind energy respectively (using MATLAB-EnergyPLAN)	Based on the simulation results, it was conclusively stated that deploying RE in countries with power plants using high natural gas share is the best measure for a rapid decrease in overall NG consumption.
Wasiq & Valerie (Ahmed and Eveloy, 2022)	United Arab Emirates (UAE)	Increasing the share of PV, with thermal, battery and hydrogen for the power sector, reverse osmosis desalination, district cooling electric transportation, and industrial electrification	Eleven (11) different scenarios consisting of different combinations of RE share and storage systems are analyzed	Their study conclusively stated that when contemplating an upper limit of 40% electrification, the electricity source should have a significant proportion that is at least partially renewable. Additionally, more flexibility should be added to allow for the usage of Excess Electricity Production produced by the additional renewable capacity needed for electrification.
Henrik et al. (Lund et al., 2022)	Denmark	Achieving full decarbonization in electricity, transportation, and industry by 2045.	Different RE technologies were modelled for different sectors in Denmark.	The main conclusion of the study is that within the constraints of Denmark's share of global sustainable biomass resources, a fully decarbonized Danish society by 2045 is technically feasible, including Denmark's part of international aviation and shipping.
Samar & Truong (Ahmed and Nguyen, 2022)	Sweden	Achieving the carbon neutral targets for Vaxjo municipality in Sweden.	Different carbon neutral scenarios were developed for meeting the RE target for the Vaxjo municipality in Sweden	In conclusion, their study stated that the modelling outcomes for the future scenarios of energy demand and supply in the Vaxjo Municipality show a rise in the final energy demand from all sectors except for transportation. While the long-term power demand can increase by up to 72% in 2050 compared to 2019 and this could drive the market for green electricity investments, the long-term heat demand can be satisfied by increasing variable costs and making minimal adjustments to the delivery system.

parameters affect the energy systems in the MENA region, therefore, for a more concise discourse, this study will focus on the North African countries in the methodology section of the paper. Also, it is of worthy note that despite the similarity in the region regarding the abundant renewable and fossil fuel resources, the situation of the energy systems in each country is unique, and this ultimately bares on the future of energy production (Gri, 2017).

3. Renewable energy potential in the MENA region

A common challenge among North African countries is the soaring energy demands: however, the energy supply situations in each country are different. Countries like Tunisia and Morocco are huge energy importers in meeting these demands. Egypt, on the other hand, is a heavy energy exporter of natural gas, however, has been importing oil lately. Countries like Algeria, and Libya are members of the Organization of petroleum exporting countries (OPEC) and are important net crude oil and natural gas exporters (Supersberger and Führer, 2011). These different energy situations in these North African countries, ultimately impact the possible energy transition and integration of new forms of energy. The unique energy system challenge in these five North African countries culminates in securing a long-term stable energy supply and generating domestic economic effects. Specifically, the challenge for Morocco, and Tunisia, is reducing the importation of fossil fuels, yet achieving stable energy production to meet energy demands. The challenge to be tackled in Egypt, Libya and Algeria involve maintaining their

energy exportation, despite the increase in energy demands in the country (Supersberger and Führer, 2011). These energy challenges in North Africa beg for effective policy formulation.

Renewable energy resource is abundant in the MENA region. Solar and wind are the dominant resources (Tsikalakis et al., 2011). The direct normal irradiation in the region is between 2050 and 2800 kWh/m²/year. The abundant solar resource in the MENA region creates the opportunity for the production of solar heating and cooling, concentrated solar power (CSP), and concentrated photovoltaic (CPV). Also, the high global solar irradiation levels in countries like Lebanon, and Egypt, make for suitable conditions for photovoltaic installations, as the range is between 1920 kWh/m²/year and 2450 kWh/m²/year (IRENA, 2016). The wind resource in the MENA region is more country-specific, as countries like Morocco, Saudi Arabia, and Egypt are endowed with high wind potential, often exceeding 7 m/s. According to reports by IRENA, the Gulf countries in the MENA region have technical potential in grid-connected and off-grid wind energy of about 65 GW, and 34 GW respectively. Additionally, the technical potential of grid-connected and off-grid solar PV is about 505 GW. Other renewable energy resources which have a high potential for exploitation in the MENA region are hydropower, geothermal, and biomass (Hawila et al., 2014b). The potential of hydropower in the MENA region is minimal, with only Iran, Egypt, Tunisia, Syria, and Lebanon having significant hydropower installations. More importantly, pumped hydro storage which plays a key role in storage capacity that can support the intermittent nature of solar and wind (I. H. Association, 2019). Geothermal

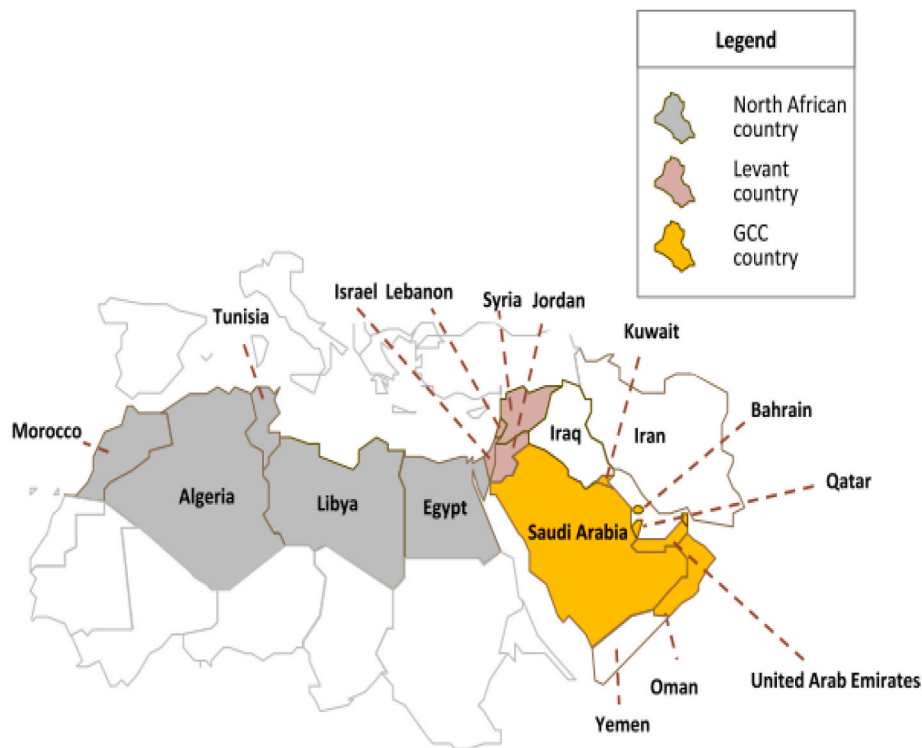


Fig. 2. MENA region's countries.

resources play a minimal role in MENA energy systems. The geothermal sites in the MENA region are found in Saudi Arabia, Algeria, and Morocco. The geothermal temperature in these countries exceeds 200 °C at 500m. This is enough potential for power applications, district cooling and desalination (Loutatidou and Arafat, 2015). Fig. 3 shows that North Africa has around 473.178 TWh of renewable energy electricity generation. Fig. 3 also shows the percentage increase between 2000 and 2019. It is seen that North and West Africa have the least increase in renewable energy integration. The highest contributor to renewable energy electricity generation in North Africa is Egypt, as shown in Fig. 4.

The following section will describe the energy situation in the five North African countries considered in this study.

3.1. Algeria

Energy transition in Algeria is a crucial issue, as the country is heavily reliant on fossil fuels for its energy production. Algeria's export comprises 93.6% of oil and natural gas, and this in itself is a challenge to the economy due to the dwindling price of crude oil in the global market.

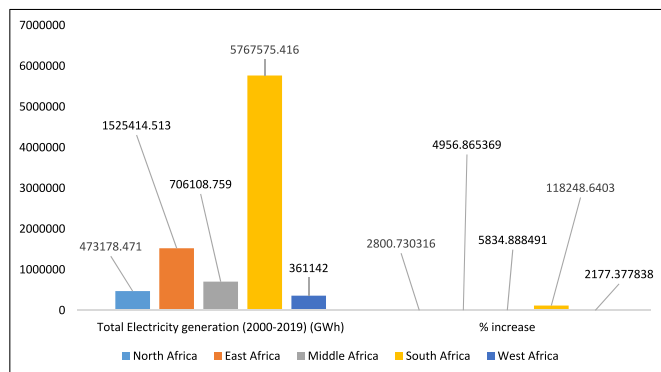


Fig. 3. Total electricity generation from renewable energy resources in regions in Africa (2000–2019) (IRENA, 2021).

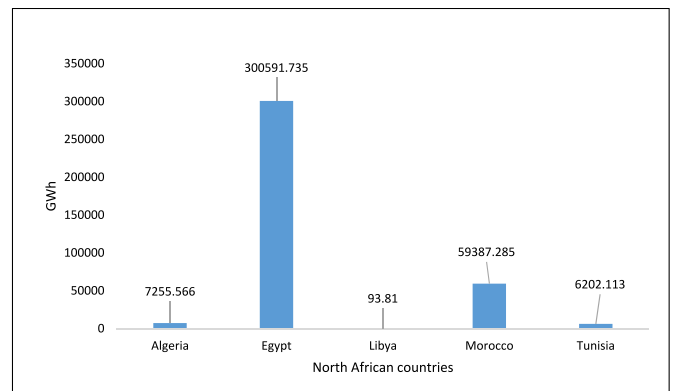


Fig. 4. Renewable electricity generation between 2000 and 2019 in North Africa.

Algeria is among the top three producers of oil in Africa (KhraiefShahbaz et al., 2018). According to the Statistical Review of World Energy (S. R. of W. Energy, 2019), Algeria is ranked as the third most significant emitter of CO₂ in Africa, after South Africa and Egypt. Algeria is planning a shift towards renewable energy production, by utilizing its vast renewable resource of solar, wind, hydropower and geothermal source. The vast presence of solar resources in the southern region affords Algeria the pathway to utilize solar energy applications like photovoltaic and concentrated solar power. The mean daily sunshine duration in Algeria is 7.26 h, and the average sunshine duration is 2650 h/year. It is also estimated that Algeria receives average solar energy of about 1700 kWh/m²/year (Bouraiou et al., 2020). There are variations of wind potential in Algeria, as the average Algeria surface has a wind potential of velocity higher than 5 m/s. This makes Algeria a good terrain for wind energy technology. The hydropower potential in Algeria is also high, however, it is not being exploited (Stambouli et al., 2012). It is estimated that Algeria receives about 65 billion m³ of rain. The northern region of

Algeria has some geothermal potential, as about 200 hot springs have been identified. The temperature of about one-third of the springs is above 45°C (Boudghene, 2011), while the highest recorded temperature is 98 °C in Maskhoutin. The electric potential of biomass in Algeria is more than 1700 GWh (Akbi et al., 2017). The biomass potential is been provided by agricultural and urban wastes (about 1.33 MTOE per year). Fig. 5 shows that the harvesting of solar production became significant only by 2015, while the wind energy resource is barely utilized. Hydropower represents the major renewable energy resource utilized as there is a total of 1408 GWh generation. Solar PV and wind generation capacity between 1990 and 2020 are 733 GWh and 29 GWh respectively.

3.2. Egypt

Over the decade, the energy situation in Egypt has been a challenge, predominantly due to an increase in population size. Egypt though has huge natural resource potential. Egypt has an oil and natural gas reserve of about 4189 billion barrels and 77200 billion cubic meters respectively (Africa, 2022). According to the Energy Information Administration (EIA), outside the scope of OPEC, Egypt is the largest region in the oil province. Egypt generates 90% of electricity from natural gas and oil, with the remaining coming from hydropower and solar energy (Abubakr et al., 2022). Despite the dependence on fossil fuels for energy generation in Egypt, the region has a wealth of renewable energy resources. Egypt has set a target of 20% and 42% renewable energy integration for electricity generation by 2022, and 2035 respectively. Fig. 6 shows that hydropower is the most utilized renewable energy for electricity generation in Egypt. The huge solar and wind resource in the country is still very much untapped.

3.3. Morocco

The energy situation in Morocco is a significant one, as Morocco imports about 99 per cent of its energy for transportation, electricity and heating. Morocco does not have significant fossil fuel reserves, as it imports 16% of electricity from Spain and Algeria, through shallow integration of cross-border transmission. This puts the country in a precarious position, in terms of energy security (Moore, 2017). Also, the energy demand has been increasing in Morocco (about 8% annually), and therefore the diversification of energy production is a crucial issue. The electricity consumption is increasing at an average of 6.5% annually (Moore, 2017). The solution to the energy crisis is not farfetched, as due to the geographical location of Morocco, it benefits from a vast renewable energy resource (Experiences, 2019). Morocco has made ambitious plans to increase the power generation of wind, solar, and hydroelectric power from 9 per cent to 27 per cent by 2020, or 42 per cent. Fig. 7 shows that hydropower and wind energy resources constitute the significant renewable electricity generation in Morocco, with 11177 GWh, and 8040 GWh respectively.

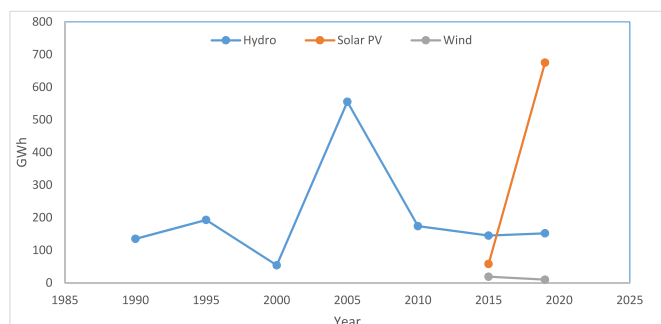


Fig. 5. Renewable energy generation by Source in Algeria (IEA, 2022).

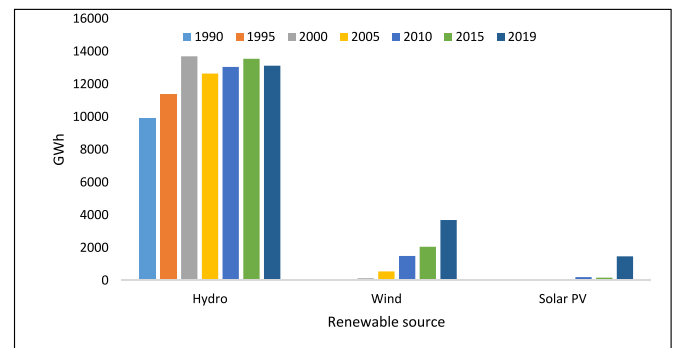


Fig. 6. Renewable electricity generation in Egypt between 2000 and 2019.

3.4. Libya

Power generation has consistently lagged behind demand for electricity due to rising demand, outdated grid equipment, a lack of periodic maintenance, and the severe damage inflicted on the electrical grid infrastructure, leading to frequent blackouts (Hafner et al., 2012). The lack of electricity in Libya was somewhat alleviated by hiring large diesel generators and importing electricity from nearby nations. At the moment, Libya imports more than 300 GWh to address the electricity shortage issue (Almaktar and Shaaban, 2021). One of the top oil producers in North Africa, Libya now produces 1.4 million barrels of oil per day, down from 1.68 million before the revolution in February 2011. About 47.1 billion barrels of proven reserves of crude oil are available. Libya has a natural gas reserve of about 52.8 cubic meters (Asheibi and Khalil, 2015). Libya has a wealth of solar resources due to its geographical location, as the average solar radiation is about 7.5 kWh/m²/day with approximately 3000–3500 sunshine hours annually (Khalil and Asheibi, 2016). Fig. 8 shows the electricity generation by source in Libya, and it is seen that solar generation is only a little percentage of the total production. Oil is the predominant electricity generation source, with 93014 GWh, while natural gas production is 70502 GWh.

3.5. Tunisia

Natural gas and petroleum reserves are the main energy resources in Tunisia. The nation produces about 87,404.63 million cubic feet (MMcf) of natural gas yearly and imports the rest from Algeria. Tunisia also imports most of its oil from Libya (Attig-bahar et al., 2021). This makes Tunisia a net importer of energy, which is due to the increasing energy demand. Tunisia's geographical location makes it have a huge wind and solar resource potential. However, there is also potential for biomass, hydropower, and geothermal. Tunisia is rich in solar resources, as the solar irradiation is more than 2300 kWh/m². The southern part of Tunisia has an abundant solar resource, with average direct normal irradiation (DNI) of 2400 kWh/m². The presence of such a huge solar resource makes Tunisia very suitable for CSP plants. Apart from the solar resource for CSP applications, the availability of a large area of land is a requirement, especially for the deployment of storage components (Zhao et al., 2018). Considering that the southern region of Tunisia is semi-Arid, and arid, with an abundant land area (desert), the country is prime for large CSP utilization. Additionally, the lands do have slopes of less than 2% which is also a net positive for CSP applications (Balgouthi et al., 2016). Favourable wind conditions are found on the north coast of Tunisia. According to the global wind atlas (IRENA, 2019), the wind speed at 80m altitudes in the North West region of Tunisia is between 6 and 7 m/s and about 8 m/s in the south. About 0.1% of the hydropower potential in Tunisia has been utilized in the country, with an installation of 72 MW for electricity generation (STEG, 2020). The share of renewable energy in current electricity generation is 3%. Fig. 9 shows that

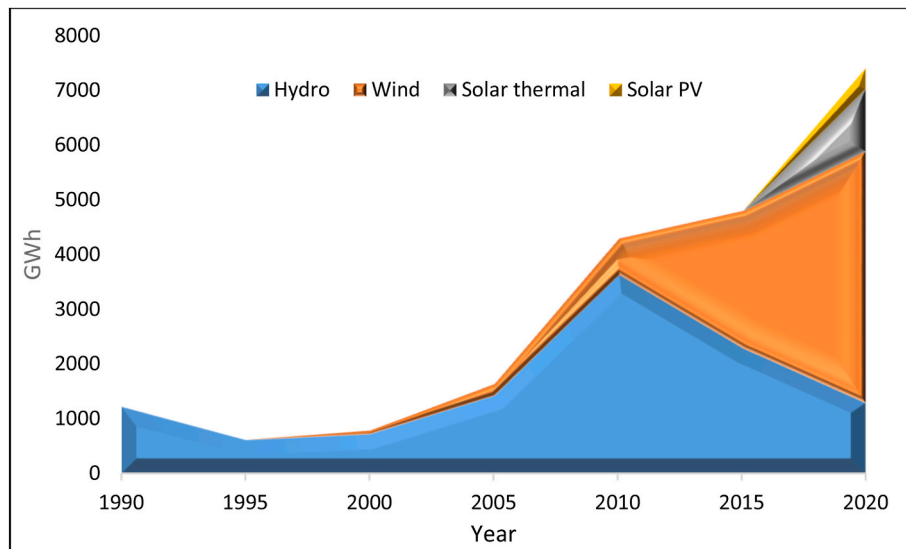


Fig. 7. Renewable electricity generation in Morocco (1990–2019).

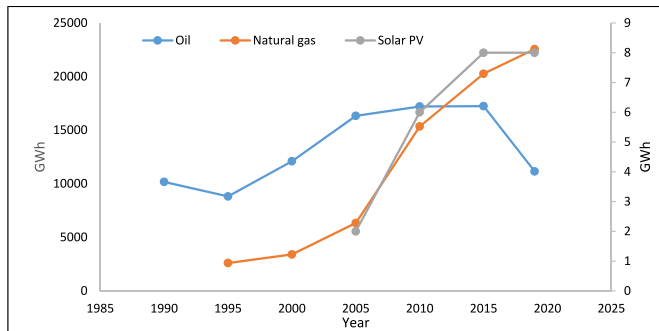


Fig. 8. Electricity generation by Source (2009–2019).

wind energy has the biggest share with a total installation of 1152 GWh, while hydropower and solar PV have installations of 479 GWh and 311 GWh respectively. Fig. 9 also shows Egypt has the highest renewable electricity generation over the past 10 years with a generation capacity of 30.06 TWh.

4. Literature review

Renewables are now seen as a key answer to the MENA region's energy and development concerns by both national governments and

foreign funders. Additionally, mounting evidence that investments in renewable energy technologies can pave the way for inclusive and environmentally friendly growth through the creation of jobs, the expansion of the private sector, the greening of the industrial sector, and environmental sustainability has piqued their interest. A study by (Aghahosseini et al., 2020) modelled a 100% renewable energy electricity scenario for MENA, considering the utilization of the vast solar and wind energy resource in the region. Their study also considered storage facilities' desalination and the industrial gas sector. A key detail in their simulation was the assumption of an interconnected grid between the 19 MENA states. Their study conclusively showed that solar and wind energy have about 90% generation capacity in all their considered scenarios. Also, it was found in their study that the estimated least cost of energy (LCOE) was around 40.3 and 52.8 Euros/MWh, based on each specific scenario. The technical feasibility of hydropower and pumped hydro storage application in MENA region was analyzed using ENERGYPLAN by (Alnaqbi et al., 2022). Their study considered the hydropower potential in the region. Their analysis showed that the infeasibility of building pumped hydro storage plants is not only hampered by the region's location and the supply of water, but as well as by the huge capital needed to plan, carry out, and run such projects. Also, it was mentioned that despite the availability of a huge supply of water in countries like Ethiopia, Egypt, and Sudan, the challenge of financial and institutional development hinders optimal hydropower development. Some studies have also investigated the utilization of

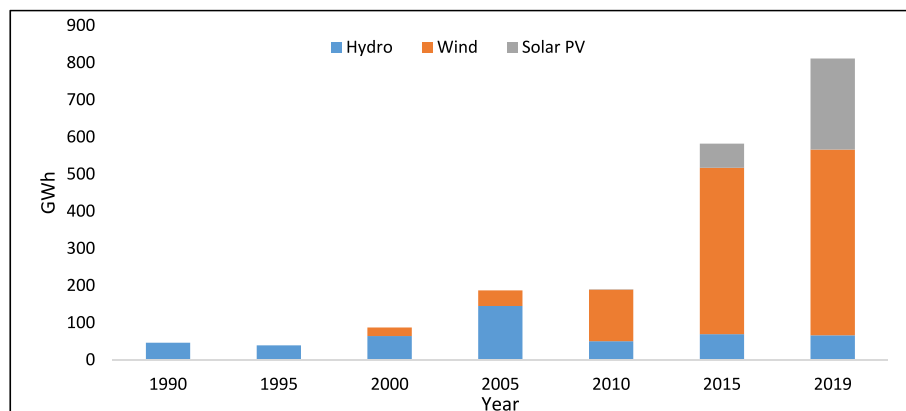


Fig. 9. Renewable energy by source in Tunisia from 1990 to 2019 (IEA, 2020).

renewable energy in specific countries in MENA region. The potential and development of renewable energy in Algeria was analyzed by (Bouraiou et al., 2020). Their study demonstrated that, despite a large potential of renewable energy, particularly solar energy, the installed capacity of both photovoltaic and wind power plants, which reached 354.3 MW (as of June 2018), as compared to the declared government targets, was a pointer to the slow integration progress of renewable energy. The study by (Souissi, 2021a) investigated the optimal techno-economic situation for grid-connected renewable energy integration in 26 locations in Tunisia. Their result showed that Wind energy was more profitable in north and central Tunisia, while it was most suitable in other locations in Tunisia. Also, their study conclusively stated that PV installations had more potential for development in southern Tunisia, with exception of Sfax, where wind energy was more economical.

The core of literature regarding the discourse of renewable energy integration in MENA countries has mostly been on the singular renewable energy resources, and on specific cities in the countries, This present study gives a more robust approach towards the modelling of the renewable energy mix in the specific country grid system. The gradual integration of high-level renewable energy potential is simulated. Also, this study bridges the gap in modelling the specific renewable energy targets in each of the considered countries, to estimate both the technical and economic feasibility of the projects. This study also brings to fore the discourse of energy-water nexus in the modelling of the different scenarios.

5. Renewable electricity targets used in this study

In this study, the modelling of the renewable energy generation for the different countries considered will be made using the targets set out to be achieved by the government. The scenarios will be made with the emphasis on the available renewable energy potential, from which the technical, economic, and environmental analysis will be simulated. In this section, a discourse of these targets will be made, and then the scenarios for analysis will be extracted.

- The Moroccan government has set a target of increasing the share of renewables in the total electricity generation to 52%, 70%, and 80% by 2030, 2040, and 2050 respectively. According to global data (Data, 2022), these lofty targets were set due to the significant progress that has been recorded in Morocco from the expansion of wind and solar projects from the initial energy transition phase. It is worth stating that Morocco only fell short of 4.9% in their target of 42% share of renewable energy capacity for 2020. Recent data put the decrease of the share of coal at 16% in the space of the last 10 years.
- The Egyptian government has been keen on integrating a sustainable energy strategy to resolve the energy crisis in the country and cultivate an environmentally sustainable and diverse electricity sector. The set target of increasing the renewable electricity share to 20%, and 42% by 2022 and 2035 respectively, is being pursued. This target is also being revised to expand it to 33% by 2025, 48% by 2030, 55% by 2035, and 61% by 2040. The set plan by the government is to have this target met through the renewable energy installation of hydropower, solar photovoltaic, concentrated solar power, and wind energy. The government believes this is an optimistic plan, and the private sector is expected to deliver most of the capacity (I. T. Administration, 2022).
- The Libyan government has set a renewable energy strategic plan, which was for the period between 2013 and 2025. This target has been expanded to 2030. The plan which was released by the Renewable Energy Authority of Libya (REAOL), was set to integrate the available renewable energy resources for power generation, thereby increasing the share of renewable energy in the electricity mix. The plan was designed to utilize solar photovoltaic,

concentrated solar power, and wind energy for meeting the set targets, which is 10% share of renewable by 2025 (L. T. Government, 2013).

- To encourage investment in renewable energy sources, Algeria has created a reverse tender system whereby power purchase agreements (PPAs) are given to businesses that offer to sell electricity for the least amount of money. A state-owned utility called Sonelgaz acts as the PPAs' counterparty. The Algerian government has set a target of 40% share of renewable energy in the energy mix by 2030 (MEI, 2021).
- Tunisia has set out two pathways for implementing its energy transition strategies, which is the increase in energy efficiency and renewable energy production. The objectives of this transition were developed in 2014, and was revised in 2019. It involves a reduction of energy consumption by 30% before 2030, and increasing the share of renewable energy by 30% in 2030. This pathway is usually referred to as 30/30 (IRENA, 2021).

6. Methodology

This section of the work gives an overview of the simulation environment used for modelling the technical, economic, and environmental feasibility of renewable electricity generation in the considered region and countries. Also, the variables utilized in the system modelling is presented alongside some assumptions.

6.1. EnergyPLAN

The EnergyPLAN software was used in this study for achieving the objectives. The EnergyPLAN is an hourly timestep simulation model for analyzing present and future energy systems of countries and regions (Østergaard, 2015). It has been used in numerous researches to model complex energy system configurations. The software utilized the bottom-up approach to resolving energy issues. The software allows for the simulation of different energy sources, both renewable and non-renewable, in a simultaneous manner. Also, the software allows for the distribution of critical excess electricity production (CEEP). The modelling of renewable energy integration in the electricity mix, takes into account the available potential in the specific region and country, using hourly time computation (Okonkwo et al., 2021). The simulation of robust storage technologies can also be done using the EnergyPLAN tool. Some of the input parameters into the software for effective modelling include electricity demand, price of carbon dioxide, fuel cost, interest rate, plant capacity, fuel consumption, and export and import electricity rate. The underlying mathematical equations used by EnergyPLAN are laid out by (Lund et al., 2014). The fundamental goal in the technical optimization process of the energy system simulation is the minimization of electricity import/export, while also utilizing the least fuel consumption solution. Furthermore, the EnergyPLAN model has high flexibility, as it can be used for simulating the energy systems for regional, industrial, transport, cooling etc., by optimizing operational strategies (Giacomo et al., 2019). This makes it a useful and effective tool for designing the feasibility of future electricity system structures, as it is done for North Africa in this study. The layout of the EnergyPLAN simulator and the optimization process used by the software is shown in Figs. 10 and 11.

The precision of the simulation environment (EnergyPLAN) is verified based on the previous studies on different regions/countries for analyzing their energy systems. Some of these studies done include Europe (Connolly et al., 2016), Nigeria (Bamisile et al., 2021), Qatar (Okonkwo et al., 2021), China (Wang et al., 2022), and Brazil (Ferreira, 2018). The scope of the analysis in literature for which the software had been utilized includes large-scale RES penetration, hybrid energy systems, and heating systems (You et al., 2018). Despite the robustness of the EnergyPLAN model, a limitation in this operation is that considering that it functions on a deterministic input-output system, it does not have the functionality of constraining input parameters by relevant

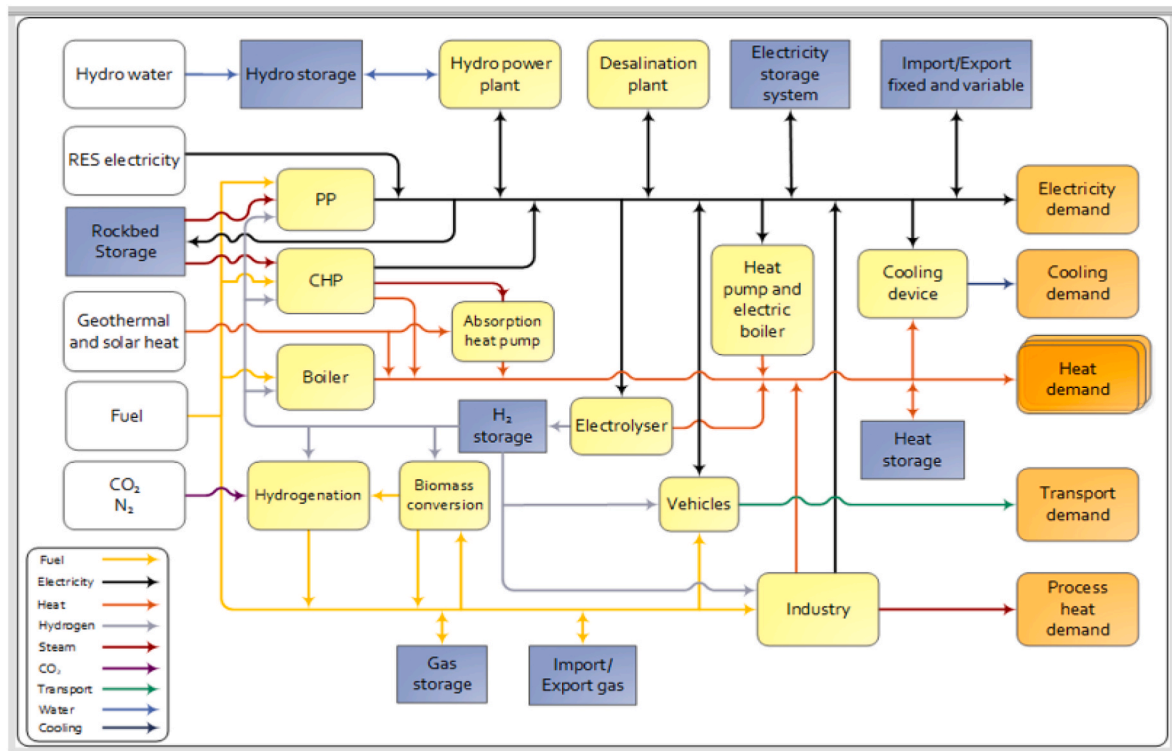


Fig. 10. Layout of energy plan simulator.

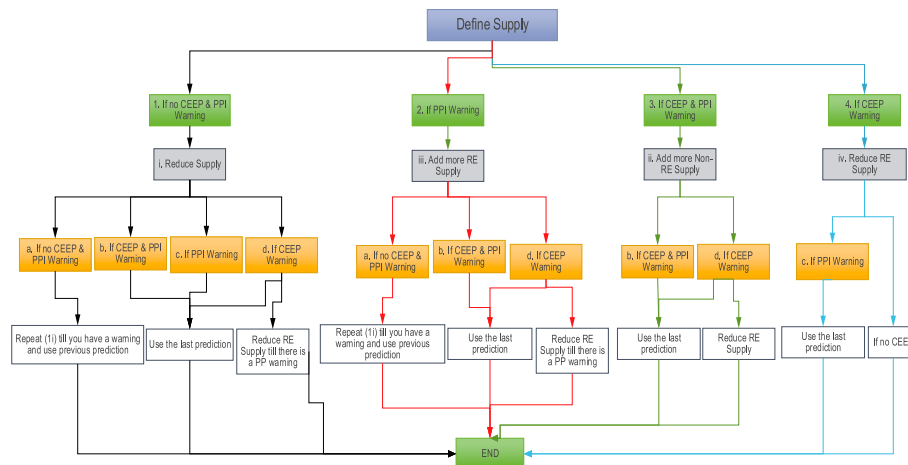


Fig. 11. Optimization simulation flowchart.

algorithms to utilize a range of input. The solution to this is by hybridizing the model with other optimization models, or by continuously adjusting the input variables, with the defined threshold of CEEP and CO₂. The summary of the research model is presented in Fig. 12.

6.2. Mathematical modelling

In this section, the mathematical modelling used in computing the technical results of the renewable energy technologies, and the economical estimations are presented. The hourly output of the Solar PV systems is calculated using equation (1).

$$P_{PV}(t) = I(t) * A * \eta_{PV} \quad (1)$$

The hourly solar insolation in kW/m² is represented using $I(t)$, the efficiency of the PV module is denoted with η_{PV} , while the PV module

area is represented using A (m²). The losses associated with the solar PV systems are also accounted for by the EnergyPLAN software.

The modelling of the wind farm is analyzed by also accounting for all the technicalities of a functional wind farm. Therefore, there is an increase in the power generation by the generators with an increase in wind speed, within the threshold of cut-in-velocity of the wind turbine. A constant rated power (P_r) is also maintained when the wind speed exceeds the rated velocity (V_r). Likewise, the generator stops its operation when the cut-out velocity is exceeded, to avoid damage. The mathematical computation of the wind turbine at hour t is estimated using equation (2).

$$P_{WT} = \begin{cases} 0 & v^3(t) - V_{ci}^3 \\ P_r & P_r \cdot \frac{v^3(t) - V_{ci}^3}{V_r^3 - V_{ci}^3} \end{cases} \quad (2)$$

For technical analysis of the hydropower, equation (3) gives the

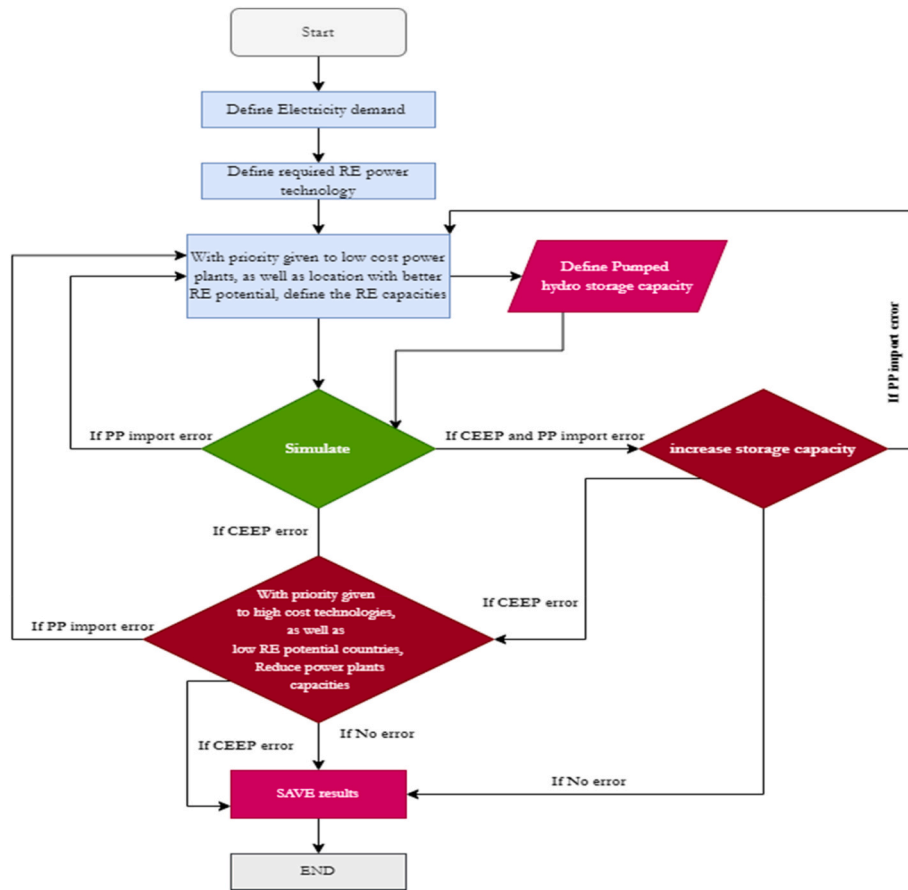


Fig. 12. Research model summary.

calculation of the dam hydropower, while the run-off river hydropower estimation is presented in equation (4).

$$P_{total} = Q^* \rho^* g^* H^* \eta \quad (3)$$

$$P_{total} = 0.5^* Q^* \rho^* \eta^* v^2 \quad (4)$$

$$Q = A \times v \quad (5)$$

Where the power output is represented as P (Watts), the density and acceleration due to gravity are shown as ρ and g , while the useable fall height is represented as h . The discharge (also referred to as flow rate) is Q , while the cross-sectional area of the channel and flow velocity are denoted as A , and V respectively.

The calculation of desalination system is calculated based on equations (6)–(9). Equation (6) gives the computation of the total electricity consumption, which is the total power of the installed pumps divided by the volume flow of fresh water. Equation (7) is the mathematical computation of the total pump power, while the installed power of the pump required to pump salt water to the desalination unit is shown in equation (8). Equation (9) gives the formula for calculating the installed power of the pumps required for the operation of the reverse osmosis desalination unit (Novosel et al., 2014).

$$E_{tot} = \frac{P_{tot}}{V_{fresh} \times 3600} \quad (6)$$

$$P_{total} = P_p + P_{RO} \quad (7)$$

$$P_p = \rho \times g \times h \times v \quad (8)$$

$$P_{RO} = c \times R \times T \times v \quad (9)$$

Where the E_{tot} represents the total electricity demand for desalination for a cubic meter of fresh water, the total installed power of the pump is represented as P_p , the installed pump power for the desalination unit is defined as P_{RO} , V_{fresh} is the volumetric flow of fresh water, v is the flow of salt water, and T is the absolute temperature.

The economic analysis in this study is focused on the total investment cost and running cost of the energy technologies. In calculating the total investment (I_{tech}) of each production, equation (10) is used

$$I_{tech} = C_{tech}^* P_{unit-tech} \quad (10)$$

Where the per-unit price and the total capacity of the technology are represented as $P_{unit-tech}$ and C_{tech} respectively.

6.3. Model validation and design portfolio

Before defining and analyzing the future optimization scenarios, a reference scenario is determined to give an in-depth picture of the current energy structure of the region under consideration. Taking the reference scenarios as the period between 2015 and 2018, the model is designed to compute the CO₂ emissions. Table 3 shows the actual electricity demand, the real CO₂ emissions, the modelled CO₂ result, and the error measured. The minimal error between the actual and model result shows the accuracy of the model developed and therefore can be used for analyzing the different case scenarios of the considered region/countries. Also, from Table 3, it is seen that the highest increase in CO₂ emissions over the years is noticed in Egypt. Also, Egypt has the highest environmental concern. This is tied to the increasing population growth and energy demand in the country.

A study by Arman et al. (Aghahosseini et al., 2020) calculated that the electricity demand in MENA is estimated to increase by 2.7%

Table 3
Model validation of the considered countries.

Year	2015	2016	2017	2018
Algeria				
CO ₂ (Actual) Mt	156.27	154.91	158.34	165.54
CO ₂ (Model) Mt	155.95	154.8	158.21	166
% deviation	-0.20478	-0.07101134	-0.0820995	0.277882525
Egypt				
CO ₂ (Actual) Mt	226.28	231.2299957	242.2299957	247.9100037
CO ₂ (Model) Mt	225.94	230	241	246.89
% deviation	-0.15026	-0.53193606	-0.50778011	-0.41144111
Libya				
CO ₂ (Actual) Mt	51.33	50.95000076	55.15999985	56.61999893
CO ₂ (Model) Mt	50.84	49.99	54.5	55.9
% deviation	-0.95461	-1.88420167	-1.19651894	-1.2716336
Morocco				
CO ₂ (Actual) Mt	60.41	60.31999969	63.52000046	64.95999908
CO ₂ (Model) Mt	60.5	59.874	63.1	64.87
% deviation	0.148982	-0.73938942	-0.66120978	-0.13854539
Tunisia				
CO ₂ (Actual) Mt	30.48	29.61000061	29.95999908	29.88999939
CO ₂ (Model) Mt	30.67	29.5	29.7	29.67
% deviation	0.623361	-0.37149817	-0.86782074	-0.73603009

(Agency, 2014), while it was estimated to increase by 3% in North Africa (On et al. Africa). A study by (Platzer et al., 2016), estimated the electricity demand of North Africa, using the econometric top-down model using macroeconomic parameters such as the growth of the gross domestic product (GDP), the development of energy intensity, etc. Based on the first scenario, an increase factor of 4.3% in electricity demand was computed for North Africa. This estimate was used in this study, however with specific consideration for each country culled from literature. The range of electricity demand increase used in this study was between 3.5 and 6%. Fig. 13 gives the future energy demand estimated for each country.

In addition, the capacity factors and economic parameters utilized in the simulation are presented in Tables 4 and 5. The capacity factor for the technologies is an important parameter in the simulation of energy systems. It is the ratio between the total quantity of energy generated throughout a period and the amount of energy it would have produced at full capacity (Cherif et al., 2014). In retrieving the capacity factor for the countries, several literature were consulted, and the average values for specific countries were utilized. In the study by El-Shimy et al. (EL-Shimy, 2009), 29 different feasible sites for PV installations were considered for Egypt. Their study analyzed the different capacity factors for the sites, from which the maximum capacity factor was computed. The location which gave the highest capacity factor was Wahat Kharga. This study utilized the 33.70% optimal capacity factor. A similar analysis was done for CSP installations in different Egyptian sites by Elmaddah (2018), for which the maximum capacity factor was

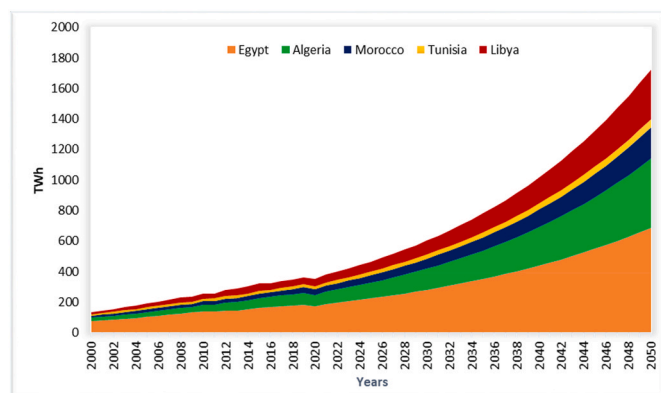


Fig. 13. Forecasted Electricity demand for North African countries considered in this study.

retrieved. The study by Dahmoun et al. (2021) numerically estimated large PV installations in different sites in Algeria. Their study showed that the optimal capacity factor was 20.64%. It is also worth stating that the capacity factor for the solar technologies retrieved from literature for each country was verified using the SolarGIS (SolarGIS) and PVsyst tools (PVsyst).

The economic parameters for the simulation process are shown in Table 5. Also, the freshwater demand used in the simulation of the desalination process in the considered countries is shown in Table 6.

7. Results and discussion

In this section, the technical, economic, and environmental implications of transitioning to renewable energy for electricity production, and for producing clean water from the desalination process will be presented. As the accuracy of the model has been validated, the results presented show a good fit for the objective set out to be achieved. Fig. 14a and b shows the solar irradiance and wind data culled from the ENERGYPLAN software.

7.1. Algeria

The Algerian economy is in a critical situation considering that about 93.7% of its exports are natural gas and oil, and the prices of these commodities have been fluctuating (Bouraiou et al., 2020). The dependence on fossil fuels for power generation has put a strain on the environmental conditions, as about 165.54 Mt of CO₂ was produced in 2018. The transition to sustainable energy development has been undertaken by the Algerian government, however, it has been a really slow process. This is despite the huge renewable potential. The results retrieved from this simulation show the estimation of the capacity of renewable energy electricity that can be generated, under the assumptions of improved technological know-how, effective policies and other factors. The result of 40% and 100% renewable energy integration into the grid system in Algeria is presented. The modelled result as presented shows that for the 40% RE integration for electricity generation, the capacity of PV, wind, CSP and Hydropower required in Algeria are 4000 MW, 5000 MW, 2000 MW, and 6000 MW respectively. The conventional power plant capacity to meet the remaining demands was 20000 MW. These installations would facilitate CO₂ emissions reduction to 37.55 Mt. This represents a reduction of 77.312% of carbon emissions, against the 2018 CO₂ emissions. According to the report by (M. of Energy), their analysis stated the CO₂ emissions avoided to be 193.3 Mt (by 2030), considering the potential of RE in Algeria. Fig. 15 shows the estimated electricity demand, and supply for 2030. The result also shows that the total annual renewable energy generation is estimated to be 55.8 TWh/yr. As of 2018, the total renewable energy electricity capacity in Algeria was 1.5 TWh (Bouraiou et al., 2020). The economic analysis of 2030 target shows that a total investment, annual and fixed cost is 49.5 Billion USD, 7.116 Billion USD, and 1.01 Billion USD respectively is required for the project. For the 100% renewable energy scenario, the estimated electricity generation from PV, wind, CSP and Hydropower were 4200 MW, 5100 MW, 2200 MW, and 6000 MW respectively. The biomass source was simulated to run on the power plants to generate additional generation.

The water scarcity in the MENA region also ripples in Algeria. In this study, renewable energy for meeting projected water demand for 2030 is simulated. The result shows that a hybrid PV, wind, and pumped hydro (Slocum et al., 2016) for the reverse osmosis desalination process will cover 40% of the total electricity requirement, on a scale of 9.8 TWh/year. The simulated PV installation, wind, and hydro to meet the 3.5 billion cubic meter (BCM) for 2030 will be 900 MW, 950 MW and 1000 MW respectively. Due to the intermittence of the RE technologies, the contribution of the conventional grid system will be 3400 MW. The estimated total investment cost for renewable energy desalination was 7.244 Billion USD, representing 2.06 USD/m³.

Table 4
Capacity factors for different technologies.

	PV	Ref.	Wind	Ref.	CSP	Ref.
Egypt	33.70	EL-Shimy (2009)	56.53	AbdelhadySBorello and Shaban (2017)	44.2	Elmaddah (2018)
Algeria	20.64	Dahmoun et al. (2021)	38	Himri et al. (2008)	35	Boudaoud et al. (2015)
Morocco	14.84		41.3	El Khchine et al. (2019)	55	
Tunisia	20	Cherif et al. (2014)	44	Souissi (2021b)	44.09	Balghouthi et al. (2016)
Libya	26	YASSER and AMDAWIALDALI (2012)	49	Elmabruk et al. (2014)	34	Ehtiwesh et al. (2019)

Table 5
Economic parameters (Pfeifer et al., 2021).

Renewable energy technology	Investment cost (MEuros/unit)	Service years	Fixed Operation and maintenance (% of investment)
Solar PV (MWe)	0.69	25	1
Onshore wind (MWe)	0.91	25	2.53
CSP	3.8	30	1.5
Pumped storage (GWh)	7.5	50	1.5

Table 6
Fresh water demand target for 2030.

Country	Projected Fresh water demand Billion cubic meter (BCM) for 2030	Reference
Algeria	3.5	Hamiche et al. (2015)
Egypt	10	Omar and Moussa (2016)
Libya	8.022	Abdudayem and Scott (2014)
Morocco	2.368	El-Ghizel et al. (2021)
Tunisia	0.491	UNDP (2013)

7.2. Egypt

The simulation result shows that Egypt will require a power plant of 39,500 MW to cater for the electricity demands for 2025, in the case scenario of conventional power generation with fossil fuel. The Egyptian government, however, have set a target of 33% renewable electricity by 2025, and 48% by 2030. The designed model was used in simulating these future targets using PV, Wind, hydropower, and CSP. The estimated installation for renewable energy technology was 8800 MW PV system, 7000 MW wind, and 9000 MW CSP. The power plant of 33000 MW was estimated to augment renewable energy production. A

reduction of 33.02% in CO₂ emission was calculated in the case of 33% RES, as against utilizing a conventional power plant in 2025. An increase in the renewable energy capacity for a 48%, and 55% share of renewable for 2030, and 2035, is shown in Fig. 16. For the 2030 and 48% renewable energy integration into the electricity grid, the critical excess electricity (CEEP) of 0.13 TWh/yr. was estimated, and this can be exported in the event of regional interconnection or to neighboring European states like Italy, or Spain. The PV, CSP, wind and Hydropower installation were measured to be 11000, 8000, 12000, and 16000 MW respectively. The total investment cost for the installation of new renewable energy technologies for this case scenario was computed as 141.260 Billion USD, while the annual investment cost was 4.3 Billion USD as shown in Fig. 18. The total annual PV electricity generated would increase from 16.51 TWh/yr. to 22.52 TWh/yr. upon increasing the share of renewable energy from 48% in 2030 to 55% in 2035. Also, the total annual wind electricity, CSP electricity, and Hydropower electricity were 64.61 TWh/year, 18.77 TWh/year, and 87.46 TWh/year respectively. For the 2040 and 61% renewable energy integration target, a 54000 MW of fossil fuel power plant was estimated to augment the 1000 MW PV, 23000 MW wind technology, 8500 MW CSP system, and 30000 MW Hydropower installations. The total investment and fixed cost estimated for the 61% renewable energy integration for Egypt were 232.93 Billion USD, and 4.67 Billion USD respectively. A 100% renewable electricity was also simulated for Egyptian electricity generation, as shown in Fig. 17. The solar PV generation was 16000 MW, while the wind and CSP generation was 30000 MW, and 11000 MW respectively. The CO₂ emission reduction for the 33%, 48%, 55%, and 61% renewable electricity targets in Egypt was estimated to be 72.4%, 73.3%, 71%, and 68.68% respectively, as against the 2018 carbon emission level.

Desalination is used mainly for drinking water supply to coastal towns and in energy sectors in Egypt. The source of the desalinated water is the Red Sea and Sinai region. The population growth has been the drive towards increasing the per capita share of fresh water in the country, and reports by UNICEF put it that the water deficit in Egypt is around 7 BCM, and there could be a lack of water by 2025. Renewable

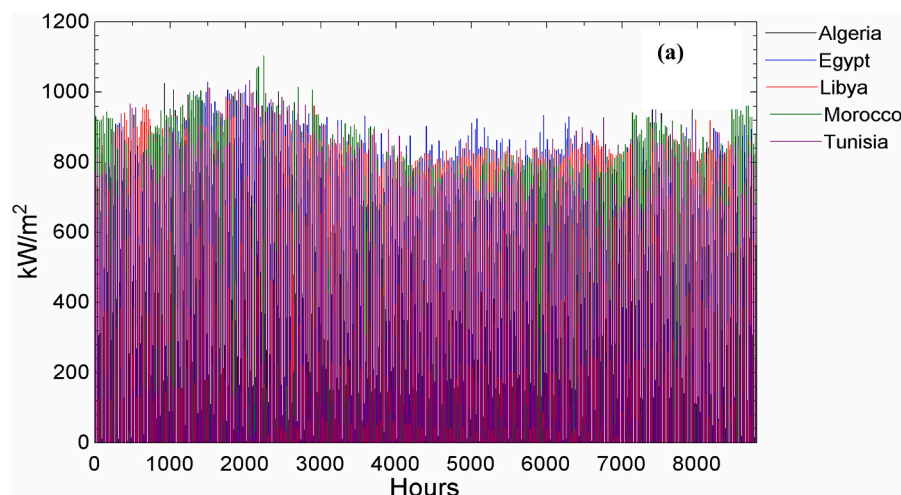


Fig. 14a. Solar irradiance data for North African countries.

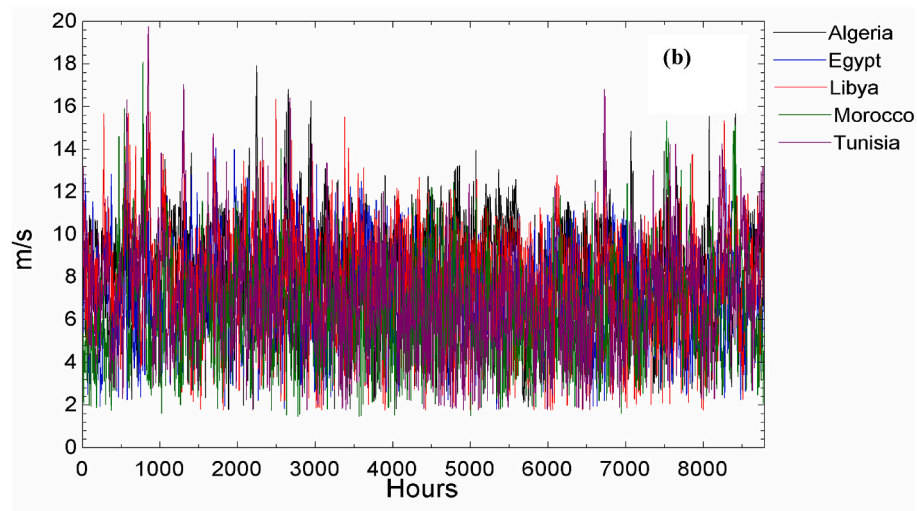


Fig. 14b. Wind data for North African countries.

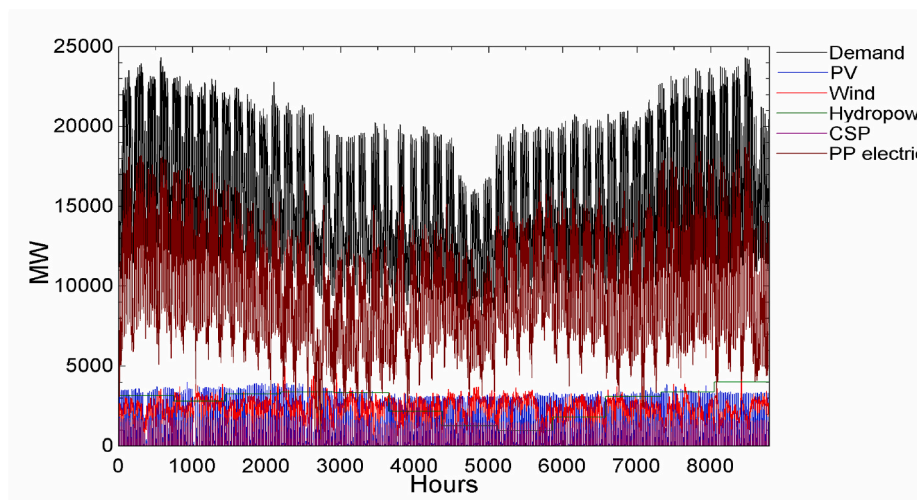


Fig. 15. Hourly electricity demand and supply using power plants and renewable for 40% renewable integration.

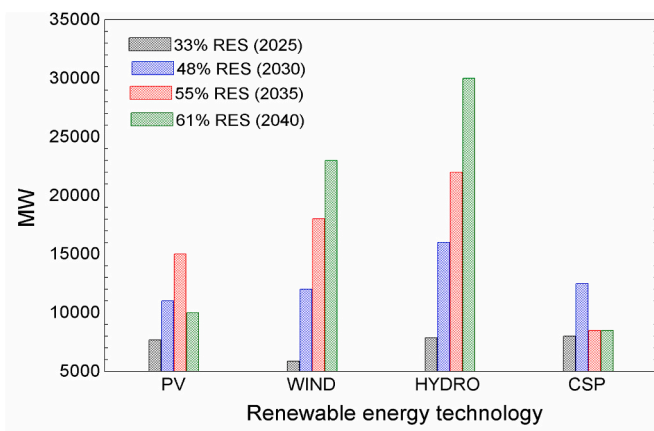


Fig. 16. Capacity of renewable energy technology for a different share of renewable energy in total electricity generation.

desalination is a viable option for Egypt based on its vast renewable resource. This simulation of the desalination in this study utilized input parameters of future freshwater demand, type of desalination process,

and the electricity demand for desalination. This study considered the RO desalination of sea water in Egypt (with a specific energy of about 6.5–9 kWh/m³) (Abou et al., 2001). The result shows that for 48% renewable electricity for desalination in Egypt, the total cost of investment was 27.2 Billion USD. For the considered year 2040, this will amount to approximately 2.7USD/m³ unit production cost. The total electricity demand for the desalination plant was computed to be 70 TWh/yr. with PV, and Wind technologies contributing 2800 MW, and 2500 MW respectively.

7.3. Morocco

The forecasted electricity demand of Morocco is shown in Fig. 19. The maximum electricity demand was seen to be 11481 MW, 20286 MW, and 35835 MW for 2030, 2040, and 2050 forecast years respectively. Based on the hourly forecast demands, the electricity generation from the renewable integrated into the current fossil fuel generation plant in Morocco is simulated. The renewable percentage in the grid which is simulated is based on the targets set by the countries. The economic simulation is based on the installation of the new components.

The simulation result estimated a production capacity of 11,500 MW of fossil fuel power plant to meet the 2030 electricity demand in Morocco. The carbon emission outcome of this scenario will be 29.661

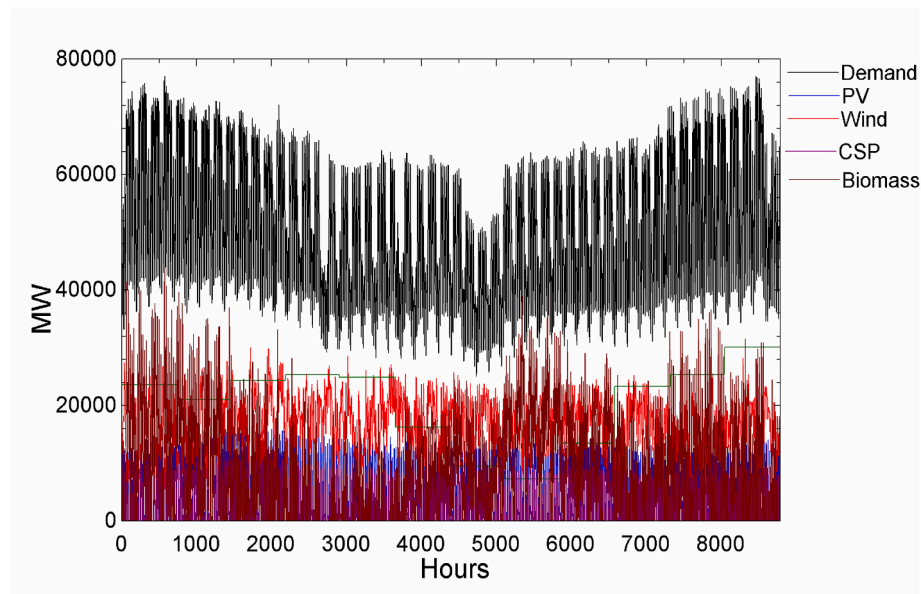


Fig. 17. Electricity demand and supply for 100% RES for 2040.

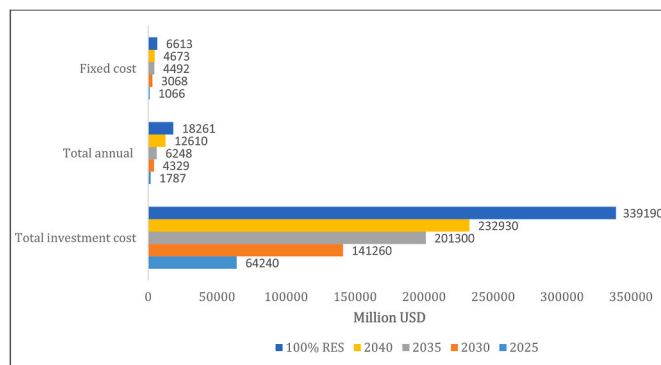


Fig. 18. Economic analysis of the renewable targets for corresponding years.

Mt. It is worthy of note that among the countries in North Africa, Morocco and Tunisia have the least fossil fuel reserves (presented in Table 1), and this has impacted the level of growth in renewable energy production and energy strategy in the country. However, the present situation of electricity generation in Morocco is that renewable energy is contributing about 34% to the total electricity generation in the country, which makes Morocco, the leading country in the region in terms of renewable energy development. Morocco, however, has set a target of 52% renewable electricity by 2030, 70% by 2040, and 80% by 2050. These targets were used in simulating the total generation from PV, wind and CSP technologies.

Fig. 20 shows the 52% renewable energy target scenario. The modelled result shows that the contribution of PV and CSP to the total generation in Morocco for 2030 will be 16.82% and 6.86% respectively. The hydropower production capacity was 4350 MW, while the Natural gas power plants will take up 9000 MW installation to meet the future demand. The total renewable electricity production was 34.15 TWh/year. The increment of the renewable share to 52% in 2030 for Morocco will cause a reduction in carbon emission by 52.09%. The emission

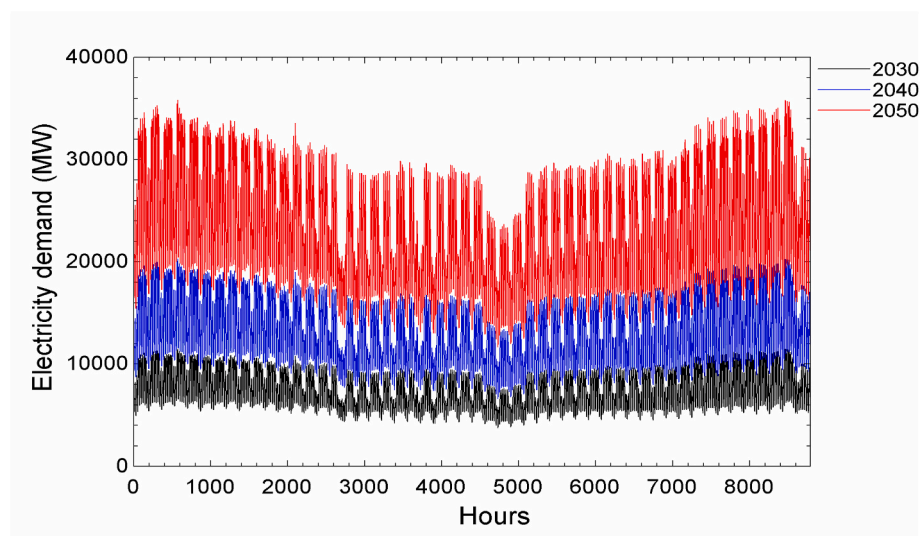


Fig. 19. Electricity demand for 2030, 2040 and 2040.

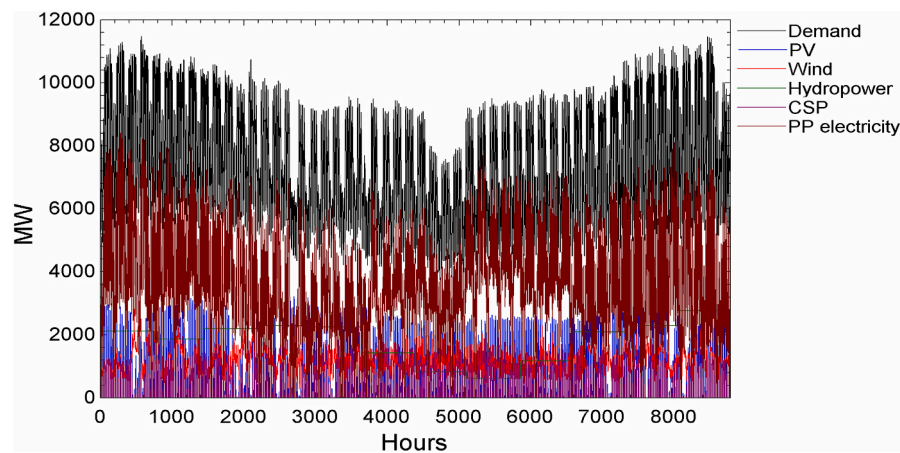


Fig. 20. Hourly electricity demand and supply using power plants and renewable for 52% renewable integration (2030).

recorded for the 2030 scenario for 52% renewable energy integration was 14.209 Mt. This represents a CO₂ reduction of 78% from the 64.59 Mt emission in 2018 (O. W. in Data). The critical excess electricity (CEEP) computed for the 52% renewable energy integration was 0.09 TWh/year, for which a storage capacity of 11 GW need to be constructed. The economic implication of such installation of renewable energy technologies was also computed, using some economic metrics like an interest rate of 1.5% (T. Economics, 2022). The total investment cost for renewable energy technology for the 52% RE integration was 37.334 Billion USD. For 2040, 70% renewable energy integration (Fig. 21 shows the 70% renewable energy target scenario), simulation results show that despite an increase in the natural gas supply, due to an increase in the electricity demand, the share of wind and solar technology (PV, wind, and CSP) was 8500 MW each, which represents a combined 147% increase in installations of solar and wind technologies. The CO₂ emission reduction for implementing this laudable project will be 74%, as against the situation in 2020. Table 7 presents the total investment and annual cost for the installation of the RE technologies to meet the RE targets for 2030, 2040, and 2050.

Morocco, is among the countries alongside Tunisia, Yemen, Lebanon and the State of Palestine, in the MENA region to pledge toward 100% renewable energy generation by 2050. Morocco, however, has not set any plan for achieving this target (Independent, 2016). The simulation result for 100% RE integration as shown in Fig. 22 show that a total peak generation of 16000 MW and 17000 MW for solar PV and wind is achieved. The total renewable electricity generation for Morocco, for 100%

Table 7

Economic and environmental analysis of the different RE integration for Morocco.

	52%	70%	80%
CO ₂ % reduction	78%	74%	64.8%
Total investment cost (M USD)	37334	87255	155350
Total Annual cost(M USD)	1118	2634	4866

RE was 213.68 TWh/year, supplied by solar PV, CSP, hydropower, wind, and biomass-run power plant. The total investment cost for this was estimated to be 177.890Billion USD.

Morocco is committed to mobilizing renewable energy to alleviate the water shortage in the country. The present state of desalination projects in Morocco can be retrieved from the report by (El-Ghizel et al., 2021). In this study, renewable energy for meeting projected water demand for 2030 is simulated. The result shows that a hybrid PV, wind, and pumped hydro (Slocum et al., 2016) for the reverse osmosis desalination process will cover 52% of the total electricity requirement, on a scale of 8.62 TWh/year. The simulated PV installation, wind, and hydro to meet the 2.368BCM for 2030 will be 900 MW, 930 MW and 1000 MW respectively. This result is within the range of the medium-term objective of Morocco for clean electricity for 2030, which was set at between 3000MW and 6000MW (Slocum et al., 2016), ("Présentation, 2020). The contribution of the conventional grid system will be 2300 MW. The estimated total investment cost for renewable

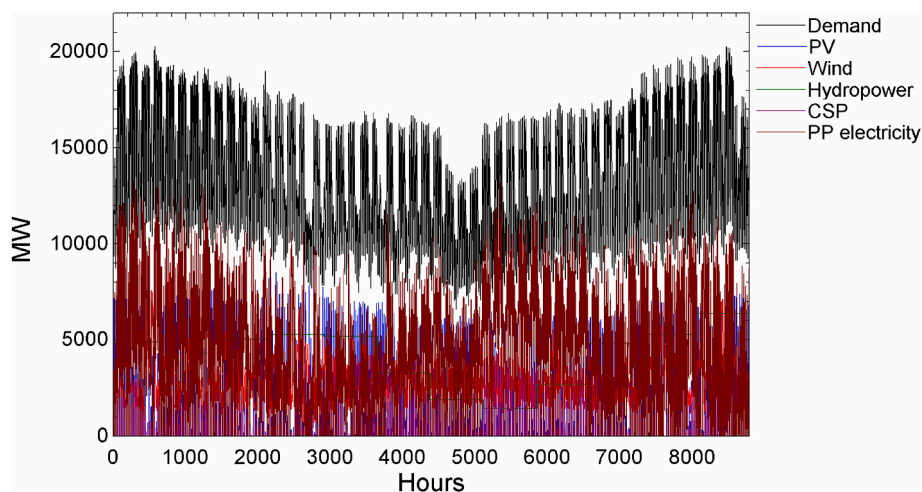


Fig. 21. Hourly electricity demand and supply using power plants and renewable for 70% renewable integration (2040).

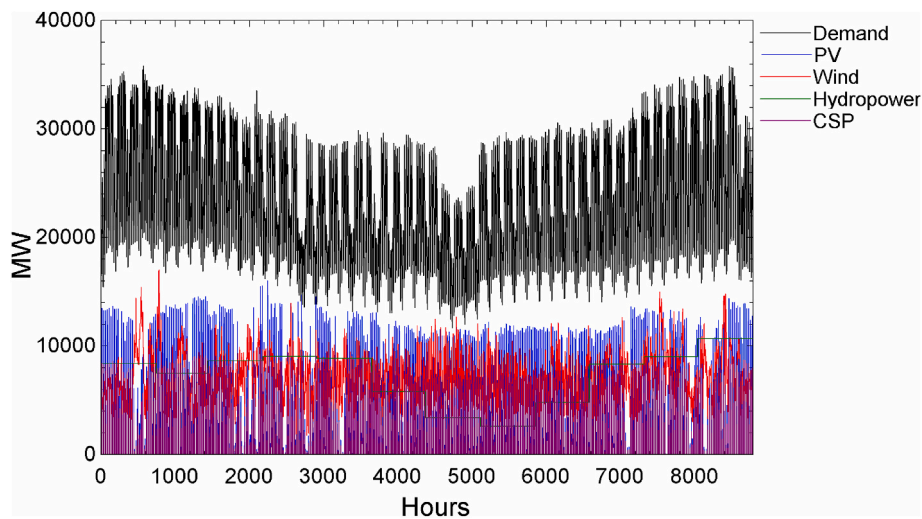


Fig. 22. Hourly electricity demand and supply using power plants and renewable for 100% renewable integration.

energy was 7.293 Billion USD. This means that the water production capacity will be 3.079 USD/m³. According to a study by Khan et al. (2018), in many countries where hybrid renewable energy technologies are used for the desalination process for freshwater production, the cost of water production ranges between US\$6.12 to US\$1.4/m³. In the case scenario of a 100% renewable desalination model, the size of the PV installation is 1800 MW. Wind energy, hydropower, and CSP technologies will contribute 2500 MW, 2500 MW, and 1500 MW respectively. Furthermore, a storage capacity of 40 GW will be required for this project.

7.4. Libya

The abundance and dependence on oil and natural gas in Libya have restricted the attention of the government to renewable energy development in the country. Exponential growth in the population has continuously placed a strain on the power sector in the country; hence, the need for diversification of electricity generation, and using renewable energy has become a necessary option. Several studies on the state of renewable energy in Libya have corroborated the huge potential of renewable and land mass for the deployment of RE technology. The context of meeting certain RE targets for future electricity demand is presented. For the projected 10% renewable electricity production for Libya, the result of the ENERGYPLAN model gives that solar PV, wind energy, hydropower and CSP can supply a combined 6.61 TWh/year. The installation of PV and wind will be 1100 MW each. In addition, as reported in the result, the economic implication showed that a total investment of 1.7 Billion USD is for the installation of the new RE technologies is required. The total annual cost is 0.479 Billion USD. In terms of the environmental analysis of the proposed RE technology for the 10% RE integration, the estimated CO₂ reduction from 2018 was 53.05%. Fig. 23 shows the electricity demand and supply for Libya for the set target of 10% renewable energy production for 2025. Fig. 23 shows that the fossil fuel power plant still supplies a huge percentage of the electricity demand, having an installed capacity of 11200 MW. However, this presents a realistic target for Libya. A lofty proposition of 100% renewable electricity was modelled for 2050, using PV, wind, CSP, Biomass and storage capacity. The installation capacity of the PV, wind and CSP, Biomass and storage capacity was 15000 MW, 25000 MW, 12000 MW, 50000 MW, and 50 GW respectively. The 100% case scenario will cost Libya 46.6 Billion USD as an investment.

One of Libya's biggest problems right now is a lack of water, and this situation will continue if no meaningful action is taken to address it. According to some assessments, Libya is one of the top countries

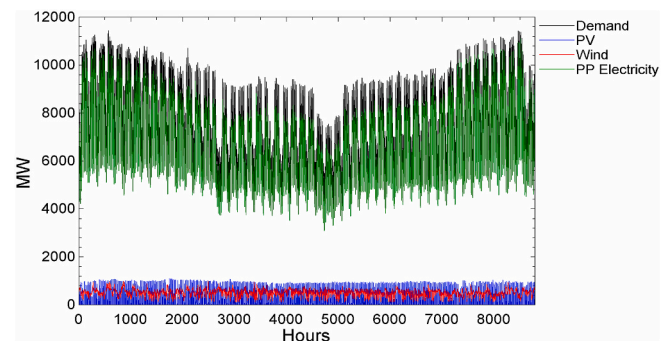


Fig. 23. Electricity demand and supply for 10% RE target with renewable technologies.

experiencing water stress, based on its water stress score of 4.84. It has consistently been rated as one of the world's most water-insecure locations (Institute, 2013). The government of Libya have not given serious attention to desalination but rather invested in man-made river project to tackle the water scarcity issue. The renewable energy desalination modelled for Libya was for 2030, and 10% renewable energy. The technologies utilized were PV, and wind technologies. The result shows that the PV, wind and fossil fuel power plant capacity was 900 MW, 860 MW, and 9600 MW respectively. Renewable energy technology will generate 5.59 TWh/year. The total investment for the RE technology for desalination was estimated at 1.4 Billion USD.

7.5. Tunisia

Since 2009, natural gas has accounted for the majority of Tunisia's primary energy consumption, reaching 52.85 per cent in 2014, followed by oil products with a 46.62 per cent share. There are no considerable energy inputs from other primary energy sources, particularly renewable energy (Brini et al., 2017).

According to the simulation results, Tunisia will need a power plant with a capacity of 4900 MW to meet its electricity needs in 2030, assuming that the BAU scenario continues. However, the Tunisian government has set a goal of 30% renewable electricity by 2030. To simulate these future targets, PV, wind, hydropower, and CSP technologies were modelled into the grid system. The installation of renewable energy technology was estimated to be 800 MW wind, 900 MW Hydropower and 1300 MW PV systems (Fig. 24). The total investment cost for the installation of new renewable energy technologies for this case

scenario was computed as 6.811 Billion USD, while the annual investment cost was 0.449 Billion USD. The total annual renewable electricity generated was 8.37 TWh/yr. Also, the total annual wind electricity, PV electricity, and Hydropower electricity were 2.52 TWh/year, 2.274 TWh/year, and 3.577 TWh/year respectively. Fig. 24 shows the electricity demand and supply for Libya for the set target of 30% renewable energy production for 2030. In terms of the environmental analysis of the proposed RE technology for the 30% RE integration, the estimated CO₂ reduction from 2018 value was 70.43%. It is worthy of note that Tunisia also has a tremendous opportunity to transfer its renewable energy, especially CSP to Europe (Italy, and Spain). A study by (Brini et al., 2017) discussed the technical feasibility of the proposed project of a regional power system between Tunisia and Europe. The proximity to Italy, resulting in less energy loss presents an ideal situation for such trade. Presently, a memorandum has been signed between the Ministry of Industry, Energy and SMEs, the Italian Ministry for the Environment, Land and Sea and the National Agency the energy Management of Tunisia (ANME) to construct an underground high voltage cable through the Mediterranean between both countries. The project is set to span 200 km and with a capacity of 1000 MW high voltage direct current. This project further solidifies the intention of Tunisia to diversify its energy generation and foster a regional power system.

The simulation for 30% RE desalination shows that a hybrid PV, wind, and pumped hydro for the reverse osmosis desalination process will cover 1.03 TWh/year of the total electricity requirement. The simulated PV installation, wind, and hydropower to meet the 491 MCM for 2030 will be 180 MW, 100 MW and 100 MW respectively. The contribution of the conventional grid system will be 550 MW. The estimated total investment cost for renewable energy was 0.948 Billion USD. This means that the water production capacity will be 1.93 USD/m³.

8. Financing RE technologies in MENA region

The financing of renewable energy projects also constitutes a significant area of discourse in the MENA region. It has been estimated that about 50 trillion dollars will have to be invested between 2015 and 2035 for global energy supply infrastructure and energy efficiency, for the world to attain the within 2 Celsius degrees of global warming (O. for E. C. and Development, 2015). Reports by REN21 (REN21, 2016) confirm that the MENA is lagging behind other parts of the globe in renewable energy investments. According to Arab Petroleum Investment cooperation (APIC), the MENA region is set to invest about \$897 billion in

energy projects, of which there is already the execution of 30% of the projects, while the 70% of total investments is budgeted for planned projects. It was also stated that the RE investment in MENA region has grown from \$0.92 Billion to \$18.64 Billion. Reports by IRENA also mentioned that Saudi Arabia, Morocco, Algeria, and the UAE are at the forefront of renewable energy investment to meet the RE targets. In Northern African region, Morocco has made the largest RE investment of about \$26 Billion. The climate investment fund (CTF) has financed the development of 960 MW of new CSP in the North African countries, with about \$0.75 Billion. Some of the supported projects are the Morocco-Noor Midelt I, and the Egypt Kom Ombo CSP. The CTF plans to invest more into future RE development in North Africa (Fund), (Nakhouda et al., 2011). The CTF has approved about 13 projects in Morocco, and Egypt, with a total of \$0.867 Billion. Apart from the CTF, there are other 11 climate funds that are available for the sponsorship of RE technologies in MENA region, towards actualizing their RE targets. Egypt, Morocco, and Tunisia, has been marked to benefit from the \$0.378 Billion program for sustainable energy financing by the Green Climate Fund (GCF). RE technology development in the MENA region are also sponsored through loans from bilateral climate finance (Nakhouda et al., 2011).

9. Renewable electricity trade in North Africa: existing and planned interconnection

In this section, the electricity interconnection among the North African countries will be explored, in the context of the development of renewable energy between the countries. The analysis will be made based on both existing and planned electrical interconnections between the countries, as shown in Fig. 25. A regional interconnection between Morocco, Algeria, and Libya was initiated in the 1950s, while a high voltage Alternating current connection between Morocco, and Spain was also built. As seen in Fig. 25, the interconnection between Tunisia and Libya is not operational since 2002. Despite the interconnections among the North African countries, the electricity trade between countries is at a modest level. A study by Kilani et al. (Ben Kilani et al., 2019) revealed that the five-line interconnection between Tunisia and Algeria, only has a transmission capacity of 1760 MW, and only 200 MW is been utilized. The cause of this modest interconnected transmission is tied to issues like institutional and infrastructural weakness at the national level, faulty regulatory framework, and issues with generation reserves margin.

The synchronous renewable energy simulation carried out will

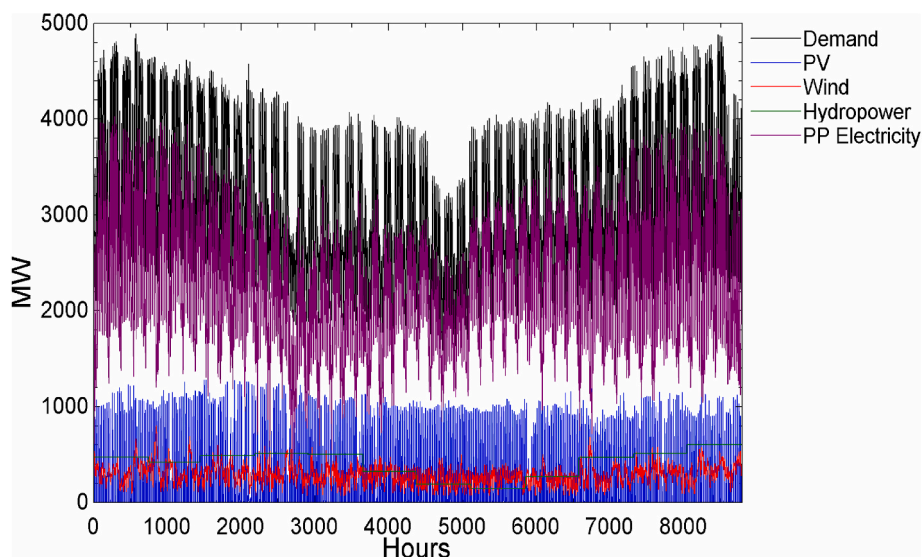


Fig. 24. Electricity demand and supply for 10% RE target with renewable technologies.

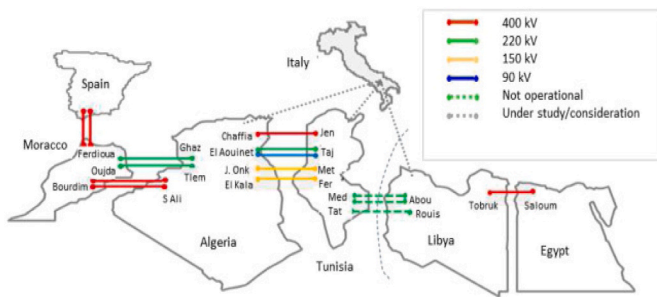


Fig. 25. Existing Electric interconnections in the North African region (Ben Kilani et al., 2019).

project development in the existing grid to accommodate the electricity generation. Fig. 26 shows the assumption of interconnected transmission grid lines, based on conceivable interconnections by 2030 (Aghahosseini et al., 2020). This study as mentioned earlier only considered the North African countries. However, this analysis can be extended to the entire MENA region.

According to estimates, the region's total annual power demand would be 602 TWh by 2030. An installed capacity of 106 GW is needed to meet this demand, which will give the area complete access to energy using just fossil fuel power plant, and this will result in annual CO₂ emissions of 273.271 Mt. Total investment expenses are 90.1 billion USD and total yearly costs are 5.71 billion USD. This study presents a novel simulation of 68% (Platzer et al., 2016) and 100% renewable energy in a synchronous grid for the entire North African region. Hydropower represents the primary source of power generation in the MENA region, while wind energy is the second largest renewable energy source in the region. Also, between 2010 and 2014, wind energy installations increased from 0.7 GW to 1.7 GW. The installation of wind energy increased to 2.5 GW, in 2017. However, the most attractive renewable energy in the region is the solar PV system. This is because of the vast potential in the region. The simulation result of the projected synchronous grid shows that for a 68% share of renewable energy, the possibility of utilizing only hydropower sources for meeting the total electricity demand of the region will require an installation of 103 GW, and 75 GW of the conventional power plant, which will cut emissions of 66.579% if only conventional PP is taken into account for grid integration while producing CEEP of 9.57 TWh. For this analysis, the five nations were taken into account. In addition, if solely wind is integrated, 96 GW wind technology will be installed, with 90 GW of conventional PP to compensate for the intermittence and peak demands. Emissions will be reduced by 66.03% if this scenario is utilized. Furthermore, Table 8 shows the result of scenarios of PV and CSP alone for the synchronous

grid systems. For the hybrid renewable technologies in meeting the 68% renewable energy projection for North Africa, the solar PV, CSP, wind, and hydropower installations were estimated to be 23 GW, 25 GW, 41 GW, and 36 GW respectively. The total renewable electricity generation with this hybrid RE integration was 410 TWh/year. The conventional power plant will contribute 75 GW generation. This will reduce emissions in the region by 67.33%, as against utilizing only conventional power plants.

These data indicate that due to the extreme intermittency of solar technology, the most probable scenario for meeting the electricity demand in North Africa through a regional transmission grid is by hybrid renewable installations across the countries. The feasibility of a synchronous grid system is only possible if hybrid renewable energy technology is utilized. The total renewable electricity generated from the combined RE systems in the region is 1723 TWh/year. In the simulation of 100% renewable electricity in the region, Biomass was included to power the power plant, and the result showed that 232 GW installation was required. Wind energy was estimated to generate 384.87 TWh/year, while hydropower was estimated to generate 279.1 TWh/year. This supports the analysis of hydropower and wind energy being the two most dominant RE in the region for meeting future demands. Furthermore, PV electricity and CSP were estimated to generate 51.5 TWh/year, and 41.3 TWh/year respectively. The total investment cost of this proposed project is estimated at 582.6 Billion USD.

The situation in Tunisia is that it has a thinly populated and unpopulated region in the sun belt ("Solar power from North Africa, 2010), therefore, the large space can be utilized for solar PV and CSP power generation to the regional grid. According to findings, nations that lack enough land availability for the generation of solar and wind energy must import their renewable electricity. Countries like Egypt and Libya have abundant land reserves with little to no trees or plants to capture solar radiation in addition to being gifted with significant solar energy resources.

Furthermore, with the continual increase in electricity demand in the MENA region, an extension of the transmission lines could be made to other MENA countries, due to land availability issues in smaller countries like UAE, and Qatar. Finally, despite that this study has not considered the Europe-North Africa renewable electricity trade, there are several such implementation possibilities. A study by ("Solar power from North Africa, 2010) presented a framework and prospects for solar trade between both continents.

10. Policy for the energy transition in North Africa

The actualization of renewable energy targets requires analysis and implementation of different policy strategies, especially for a region like North Africa, where there is a high reliance on fossil fuels for national

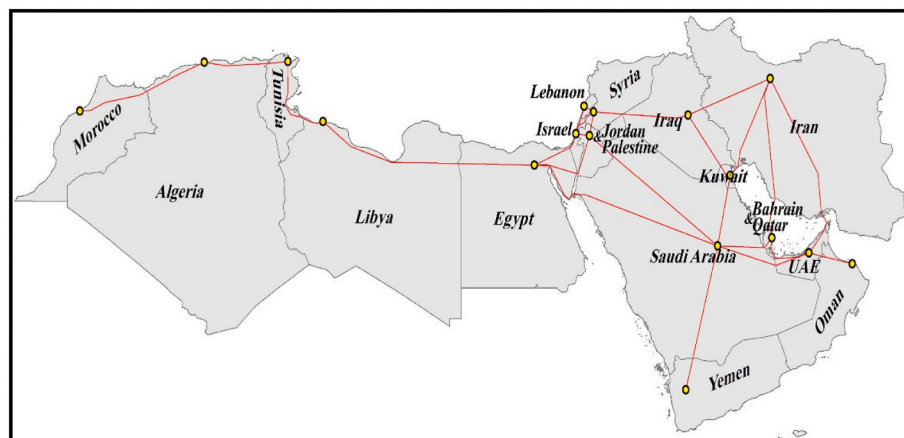


Fig. 26. Planned Electric interconnections in the North African region (Aghahosseini et al., 2020).

Table 8
Scenarios for 68% renewable energy integration for regional transmission.

Renewable technology	Installations Required	Conventional PP augmentation	% CO ₂ reduction (as against only conventional PP)	CEEP	Total Annual cost (Billion USD)
PV	199 GW	106 GW	42.52%	154.7 TWh	34.703
Wind	96 GW	90 GW	66.03%	12.68 TWh	25.48
CSP	192 GW	106 GW	42.7%	151.3 TWh	93.9
Hydropower	103 GW	75 GW	66.57%	9.57TWhh	46.85
PV + Wind + CSP + Hydropower	PV-23GW CSP-25GW Wind 41 GW Hydro-36GW	75 GW	67.33%	4.38 TWh	39.806

revenue. Effective policies will attract huge private investments, and increase society and social acceptance of renewable energy technologies. Policy considerations that could influence the next stage of renewable energy growth in North Africa majorly lie in regional cross-border cooperation on renewable energy. A continuous barrier in North Africa has been the limited amount of net imported electricity from neighboring countries, which has been a result of political barriers hindering cooperation. Policies that will ensure cross-regional renewable energy projects should be taken. More practically, the North African government should study analysis of concrete examples of countries that have embarked on cross-regional renewable energy projects. A prime example of this is the Germany-Denmark joint ground-mounted solar PV projects, and the Sweden-Norway joint electricity market project (IEA, 2020). These two laudable projects have highlighted several benefits of regional interconnections. Some of these are the benefits that Sweden is getting in terms of access to more hydropower at a very low cost, and the cheap cost of solar PV electricity in Denmark and Germany (IEA, 2020). This is very achievable in North Africa considering that there has been some grid infrastructure set up, as shown in Fig. 25. The political will to reach concise agreements among the countries will then help to tackle the technical complexities of the transmission. This has been the reason for the stalling of the Tunisia-Libya interconnection, as the infrastructure has been deemed not up to standard (Ben Kilani et al., 2019). In addition, the North African government need to leverage some policy initiatives that the European Union have made with North African states over the years. In the case of Algeria, the primarily North African gas exporter, EU member states buy a considerable share of the region's gas production and have strong negotiating power (Lacher and Kumetat, 2011). These negotiations can be diversified into renewable energy cooperation agreements between the countries. Also, the only functional transmission between North Africa and Europe (which is between Spain and Morocco), has been mainly one-sided. The 400 kV line, constructed in 1997, running through both countries, has mainly transmitted Spanish off-peak electricity to Morocco. The government of Morocco should draw up a project agreement to sell some of its renewable energy to Spain. These suggestions of course are subject to the acceptance of these options by European. Also, only a complicated and pricey new grid system would be able to export electricity to Europe (Pfeifer et al., 2021).

Also, the overall cost-competitiveness and ease of access to finance for renewables, particularly in comparison to the strenuous planning processes required for nuclear technology, could be reflected in future policy updates.

Conclusively, North Africa's energy system's evolution and transition will be difficult but inevitable. For the time being, advancements will primarily depend on transactional exploitation of simpler opportunities, especially on a bilateral basis when relations are cordial. The implementation of a consistent set of strategies, action plans, and policies aimed at not only evolving the regional energy system but also developing the energy, political, and social systems that collectively define the energy system possibilities will lead to greater success if policymakers become more aware of the integrated set of challenges and

opportunities they face.

11. Conclusion

In this study, a thorough discourse on the situation of electricity generation for power and desalination in MENA, and more in-depth in North Africa is presented. This study analyzes the present condition of electricity generation, renewable energy capacity and renewable energy targets in five North African countries (Algeria, Morocco, Tunisia, Libya and Egypt). Then this study simulates using ENERGY PLAN software, the technical, economic, and environmental outputs of set renewable energy targets in the case study countries. The existing electricity grid interconnection between the North African countries is also taken into account in proposing renewable energy trades. This simulation is made with the assumption that the grid transmission capacity will be developed to accommodate the electricity generation. Some of the significant results retrieved from this study, which can be useful in future policy development, technical decisions, and further study are:

- The increment of the renewable share to 52% in 2030 for Morocco will cause a reduction in carbon emission by 52.09%. The emission recorded for the 2030 scenario for 52% renewable energy integration was 14.209 Mt. This represents a CO₂ reduction of 78% from the 64.59 Mt emission from 2018
- The modelled result shows that for the 30% RE integration for electricity generation, the capacity of PV, wind, CSP and Hydropower required in Algeria are 4000 MW, 5000 MW, 2000 MW, and 6000 MW
- The result gotten shows that for 48% renewable electricity for desalination in Egypt, the total cost of investment was 27.2 Billion USD. For the considered the year 2040, this will amount to approximately 2.7USD/m³ unit production cost
- For the projected 10% renewable electricity production for Libya, the result of the model gives that solar PV, wind energy, hydropower and CSP can supply a combined 6.61 TWh/year. Policy considerations that could influence the next stage of renewable energy growth in North Africa majorly lie in regional cross-border cooperation on renewable energy
- The Tunisian government has set a goal of 30% renewable electricity by 2030. The installation of renewable energy technology was estimated to be 800 MW wind, 900 MW Hydropower and 1300 MW PV systems to meeting this target.
- The actualization of renewable energy targets requires analysis and implementation of different policy strategies, especially for a region like North Africa, where there is a high reliance on fossil fuels for national revenue
- The situation in Tunisia is that it has a thinly populated and unpopulated region in the sun belt ("Solar power from North Africa, 2010), therefore, the large space can be utilized for solar PV and CSP power generation for the regional grid

The authors suggest that future studies should consider a more

specific technical analysis of the interconnection infrastructure and weakness in the region. In addition, the energy-water-food nexus in North Africa especially with respect to meeting future electricity demands should be investigated.

CRedit authorship contribution statement

Humphrey Adun: Conceptualization, Methodology, Methodology, Writing – original draft. **Hamagham Peter Ishaku:** Data curation, Methodology. **Ayomide Titus Ogungbemi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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